



集成电路设计工程 2025秋

Lecture 13 上机考试样题及作业解析

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清华大学

2025年12月16日

上机考试要求



- 地点： 自强科技楼2号楼机房
- 时间： 考试周期间（TA课后会统计大家的时间）
- 考试时长： 2小时
- 题目： 模拟数字二选一
- 涵盖范围： 包含电路设计，电路仿真，版图设计，版图后仿。主要考察同学们对于集成电路工程技巧的掌握能力。

5.1、Comparator



- ◆ 拟采用65nm CMOS工艺实现如下图所示的比较器，晶体管尺寸已标注，电源电压1V，偏置电流10uA，VIN端固定输入DC电平0.5V，VIP端输入时变信号，采用analogLib库下面的vpwl信号源，所加激励信号时间点1为0s时输入0V，时间点2为2us时依然输入0V，时间点3为5us时输入渐变到1 V，时间点4为7us时输入维持1V，时间点5为10us时输入渐变到0V，OUTN和OUTP分别连接10fF的analogLib理想电容。试完成如下上机测试。



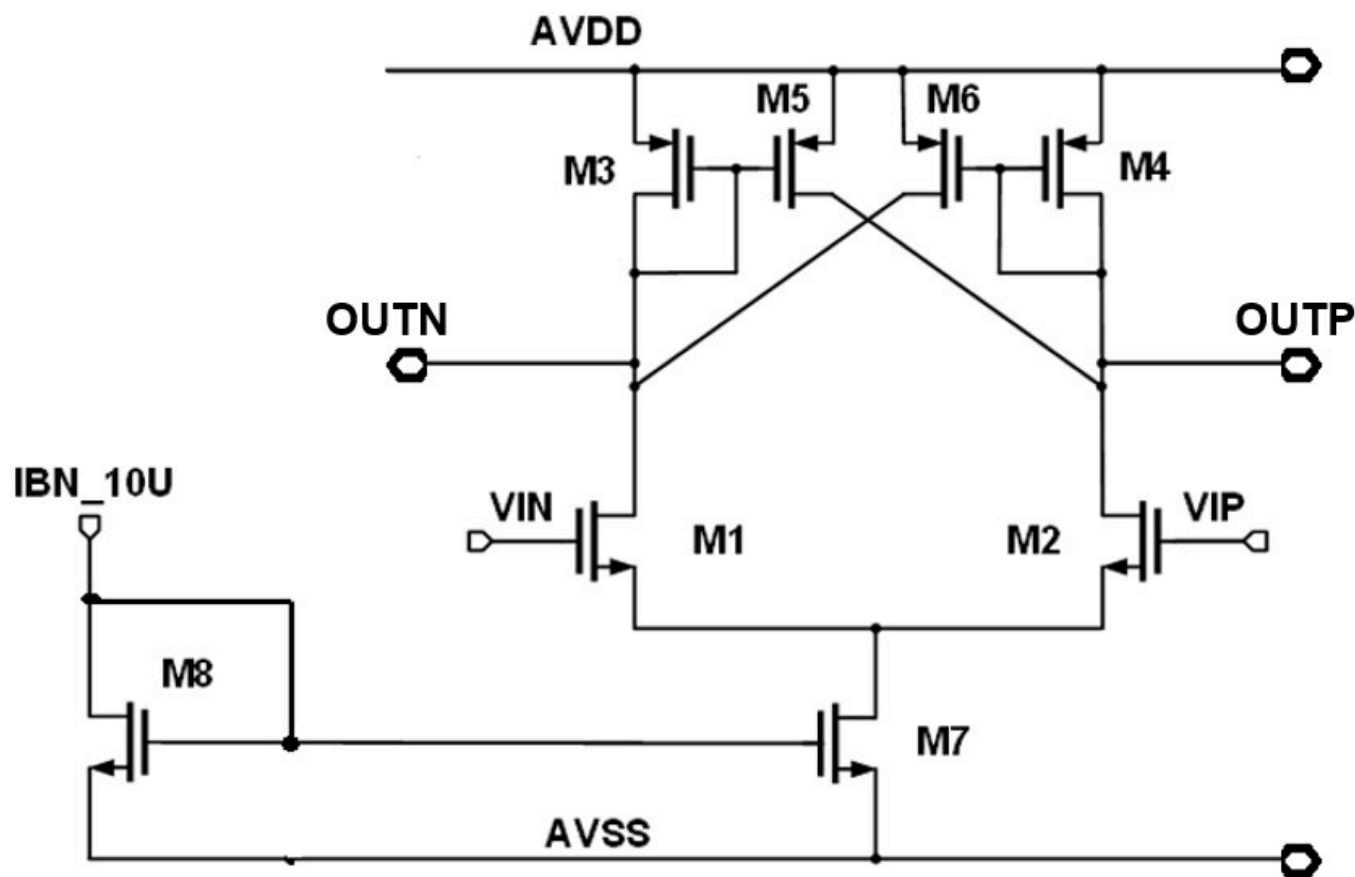
5.1、Comparator

- ◆ 要求;
- a) 在Cadence环境下完成芯片的电路图，并建立其Symbol;
- b) 按下图所示建立其测试电路图;
- c) 在ADE环境下完成瞬态仿真（10us，仿真精度errpreset设为conservative），打印输入VIP，VIN，输出OUTN以及OUTP节点的波形，并解释;
- d) 完成该芯片的版图设计，并进行DRC和LVS检查;
- e) 对所设计的版图进行寄生提取后进行后仿真，打印输入VIP，VIN，输出OUTN/OUTP节点的波形，并计算比较阈值;

5.1、Comparator



◆ Comparator的电路图如下;

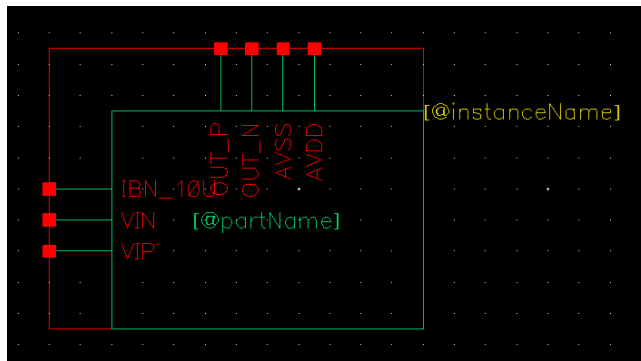
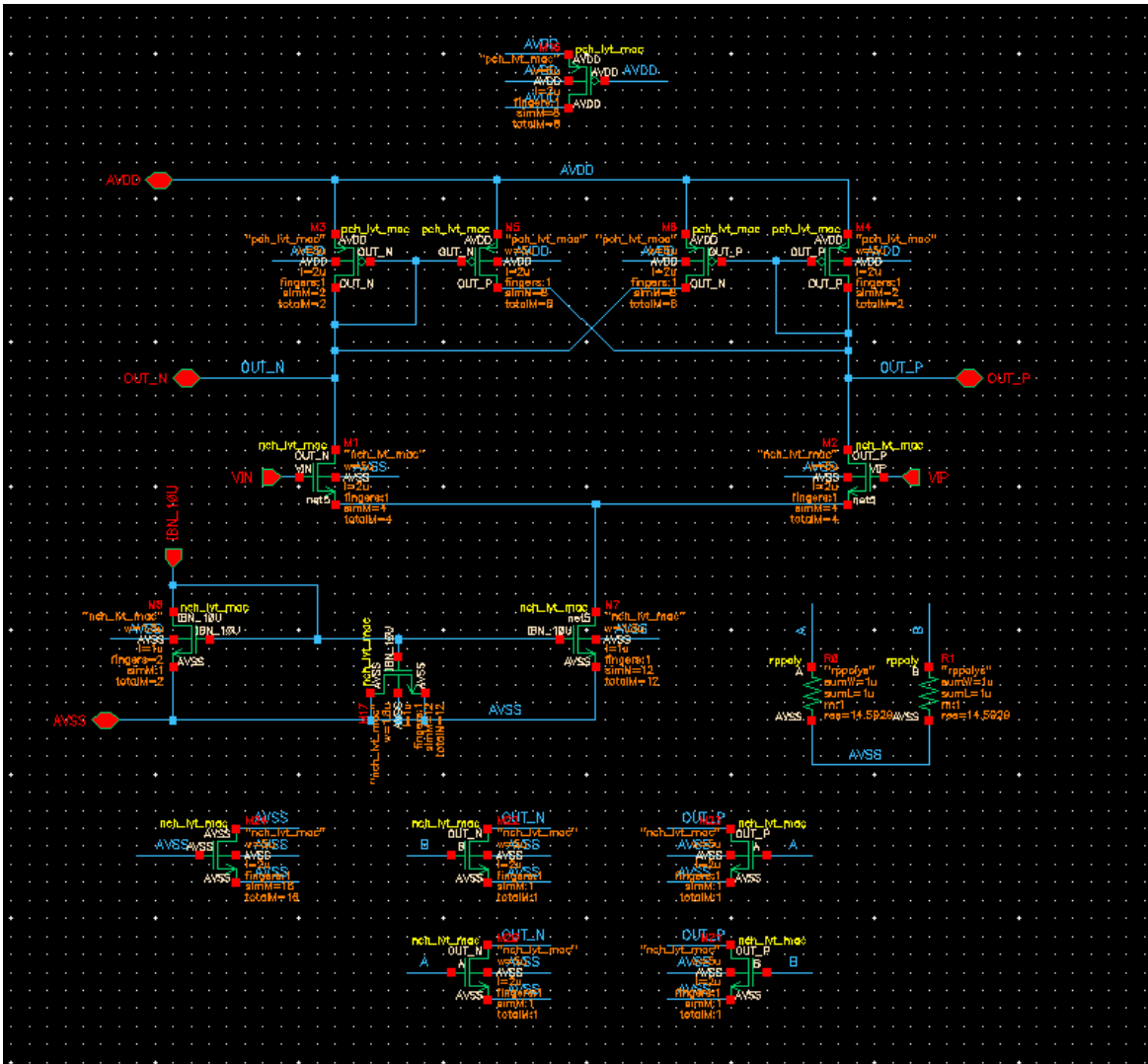


5.1、Comparator

◆ Comparator

电路搭建;

1、Comparator schematic;



2、Comparator symbol;

5.1、Comparator

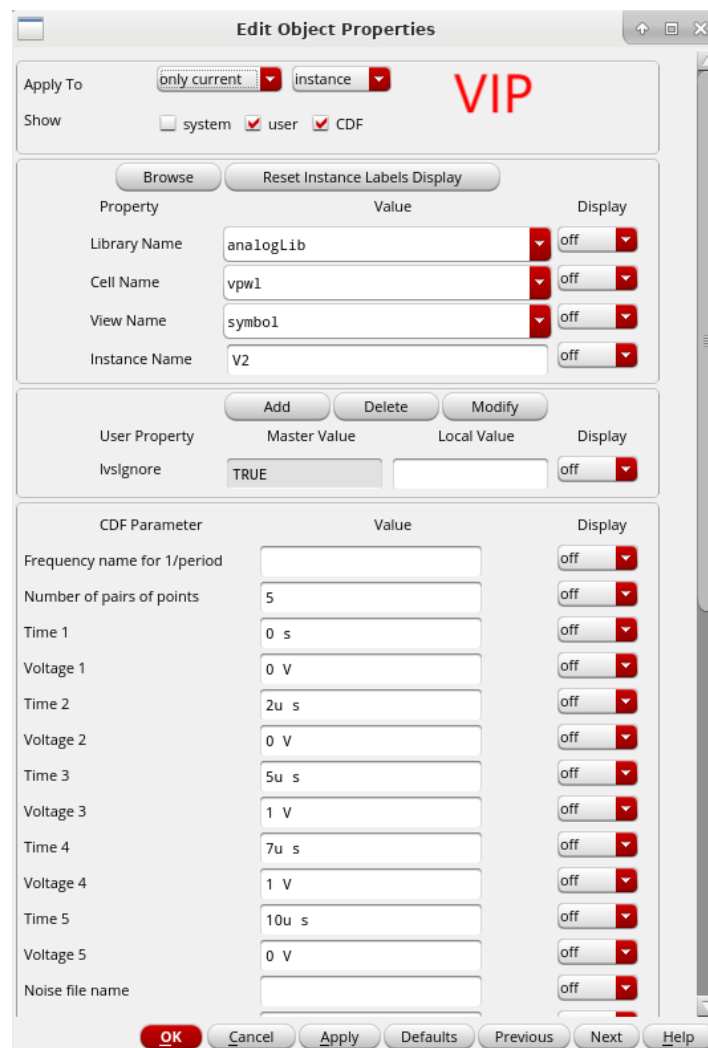
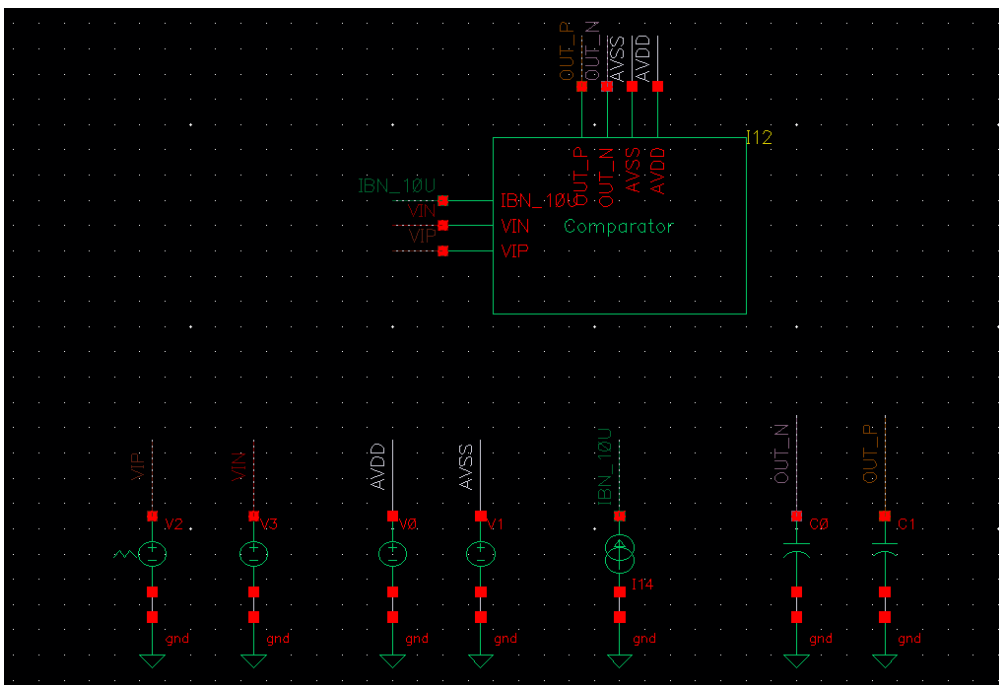
◆ Comparator电路参数;

	L/um	单个W/um	Fingers	Multipliers
M1,M2	2	5	1	4
M3,M4	2	5	1	2
M5,M6	2	5	1	8
M7	1	1.8	1	12
M8	1	1.8	1	2
M17	1	1.8	1	12
M20,M21,M22,M23	2	5	1	1
M24	2	5	1	16
R0,R1	1	1		1
M16	2	5	1	8

5.1、Comparator测试电路图



b) 按所示建立其测试电路图;



1.电源电压1V

2. 偏置电流10uA

3. VIN端固定输入DC电平0.5V

4. VIP端输入时变信号，采用analogLib库下面的vpwl信号源，时间点1为0s时输入0V；

时间点2为2us时依然输入0V；

时间点3为5us时输入渐变到1 V；

时间点4为7us时输入维持1V；

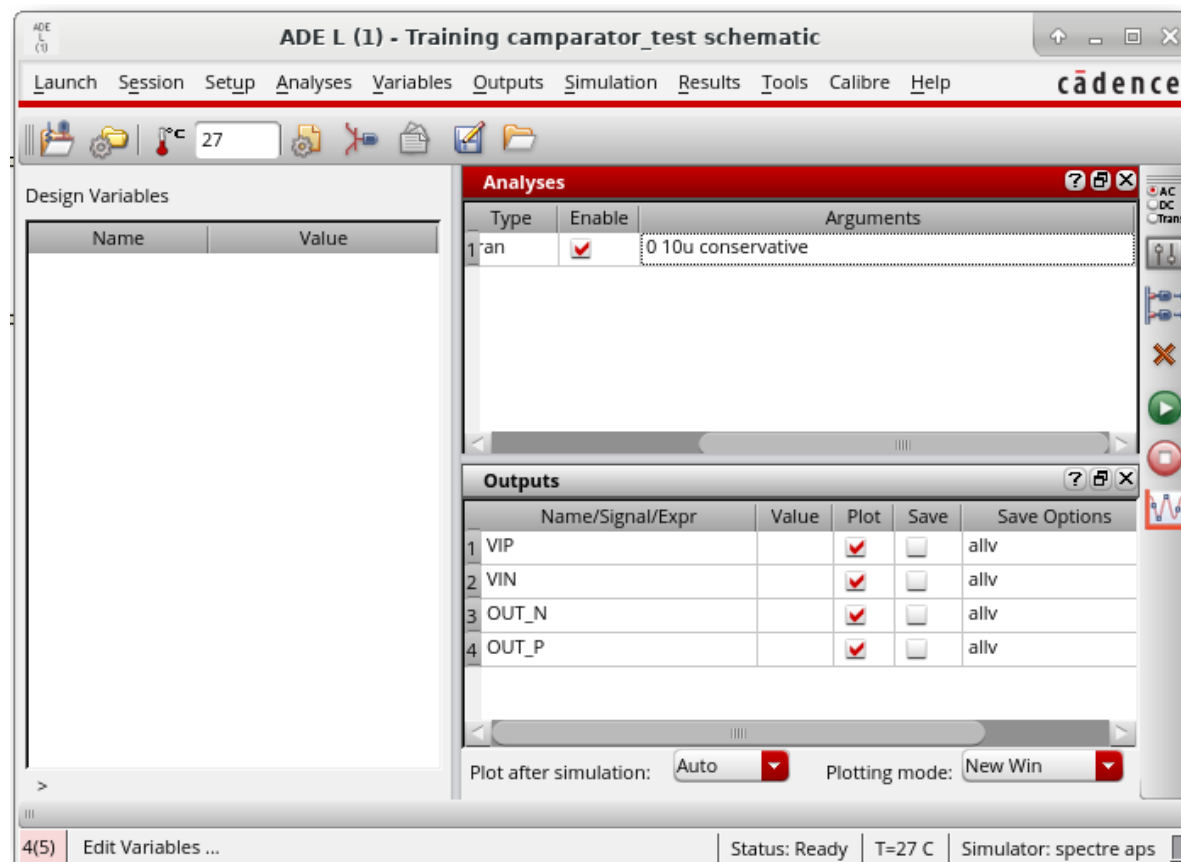
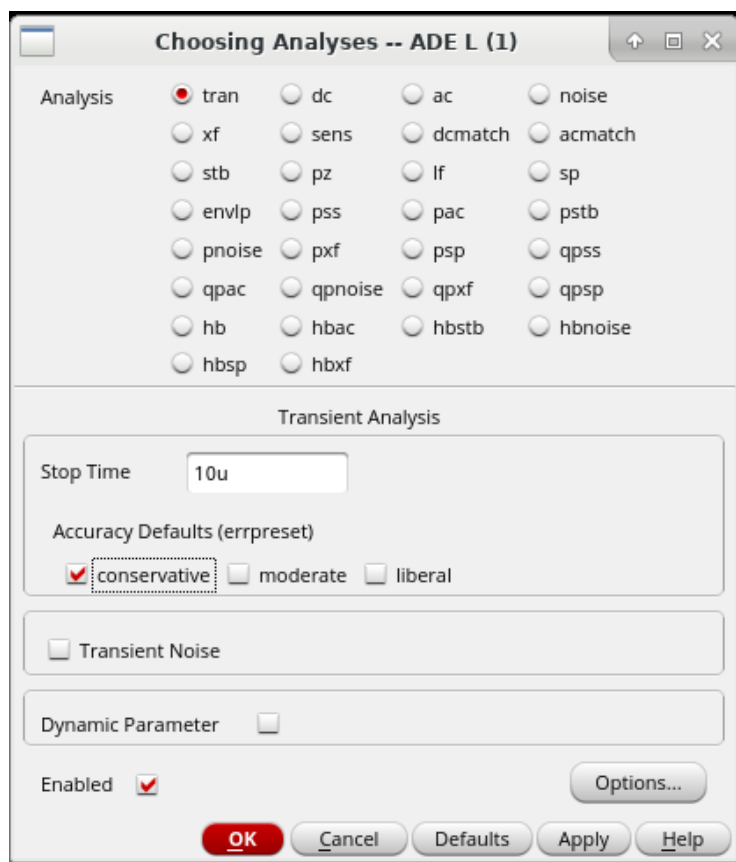
时间点5为10us时输入渐变到0V。

5. OUTN和OUTP分别连接10fF的analogLib理想电容。

5.1、Comparator瞬态仿真——前仿



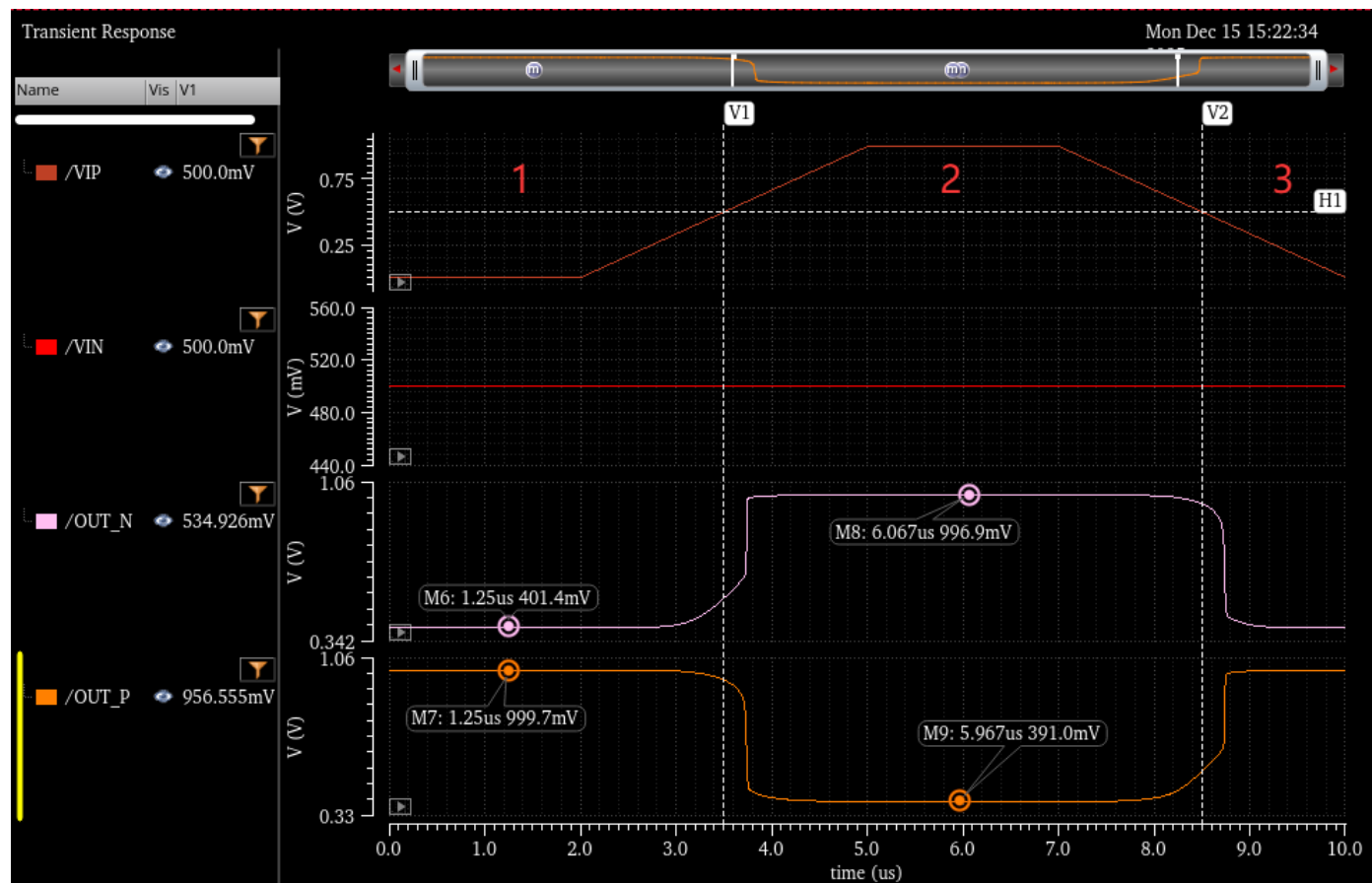
c) 在ADE环境下完成瞬态仿真（10us，仿真精度errpreset设为conservative），打印输入VIP，VIN，输出OUTN以及OUTP节点的波形，并解释；



5.1、Comparator瞬态仿真——前仿



c) 在ADE环境下完成瞬态仿真 (10us, 仿真精度errpreset设为conservative), 打印输入VIP, VIN, 输出OUTN以及OUTP节点的波形, 并解释;



第一段: $V_{IP} < V_{IN}$ 。VIP一侧基本没有电流流过, OUT_P被上拉至VDD; VIN一侧的mos管导通, $OUT_N \geq 2 \cdot V_{ov}$ 。

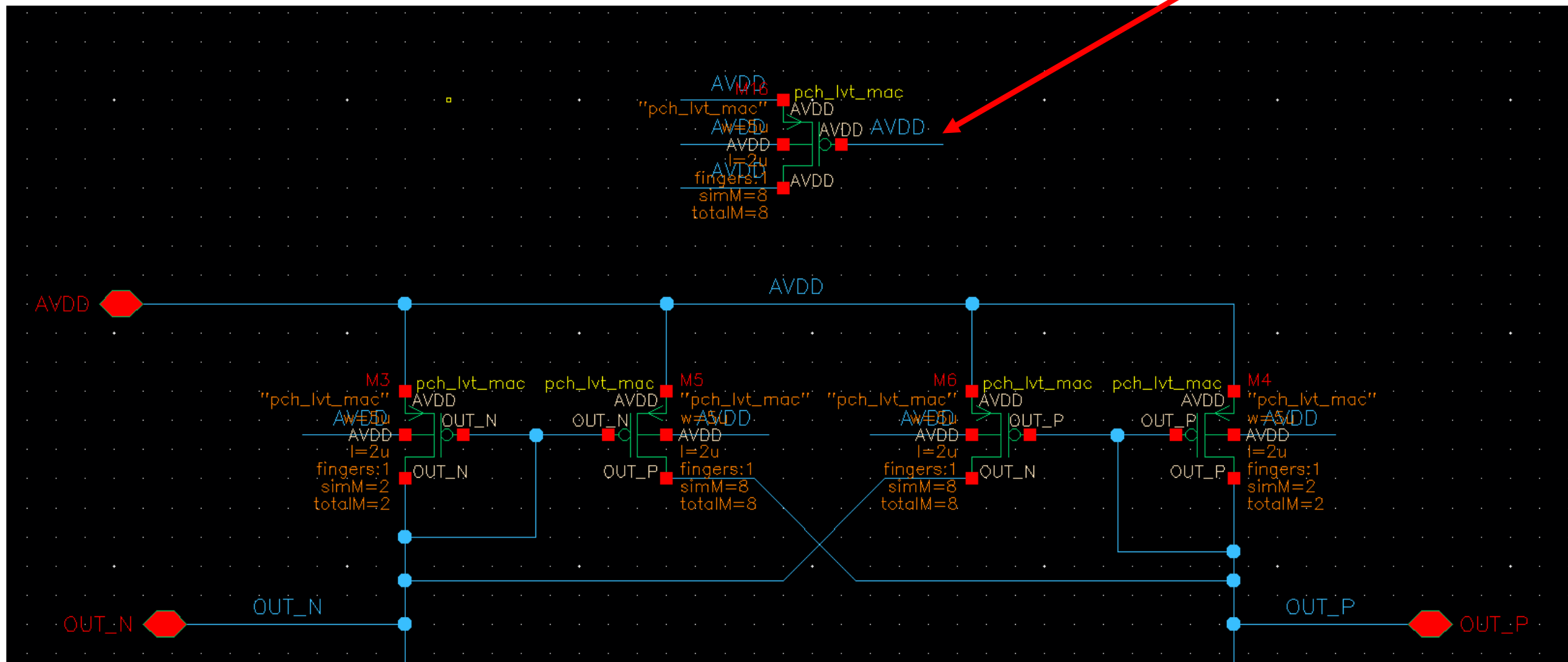
第二段: $V_{IP} > V_{IN}$ 。随着VIP的增大, VIP一侧的mos管逐渐导通, OUT_P逐渐减小, 并满足 $\geq 2 \cdot V_{ov}$; VIN一侧的电流减小至接近0, OUT_N逐渐增大至VDD。

第三段: $V_{IP} < V_{IN}$ 。与第一段类似。

5.1、Comparator



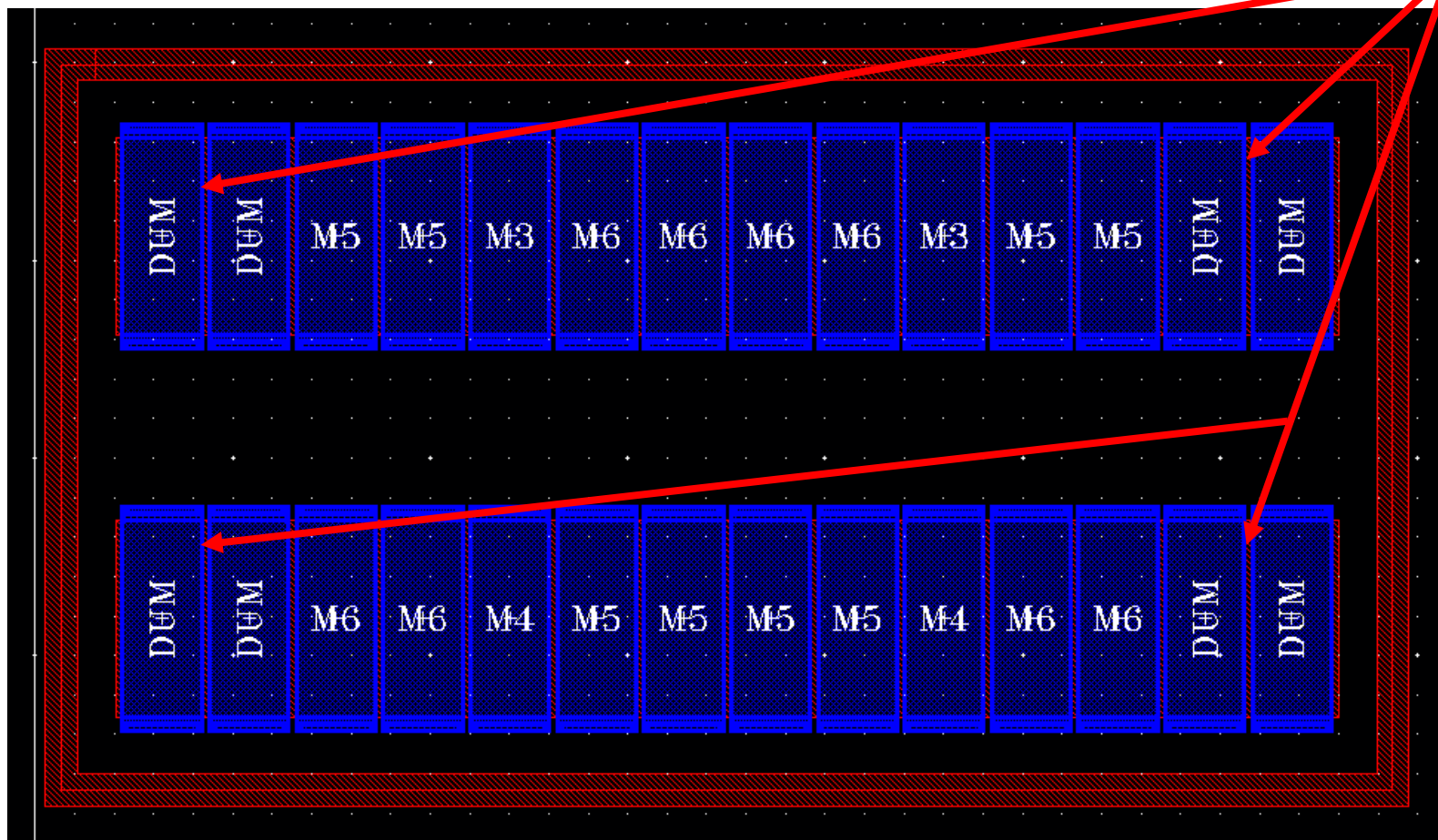
◆ Comparator版图 (M3、M4、M5、M6、M16布局) ; 1、M16为Dummy;



5.1、Comparator



◆ Comparator版图 (M3、M4、M5、M6、M16布局) ; 1、M16为Dummy;



1、如左图所示，M5、M6为ABBA二维共质心版图，M3、M4分布在M5、M6中间，呈对称分布，并在左右两边加入Dummy MOS;

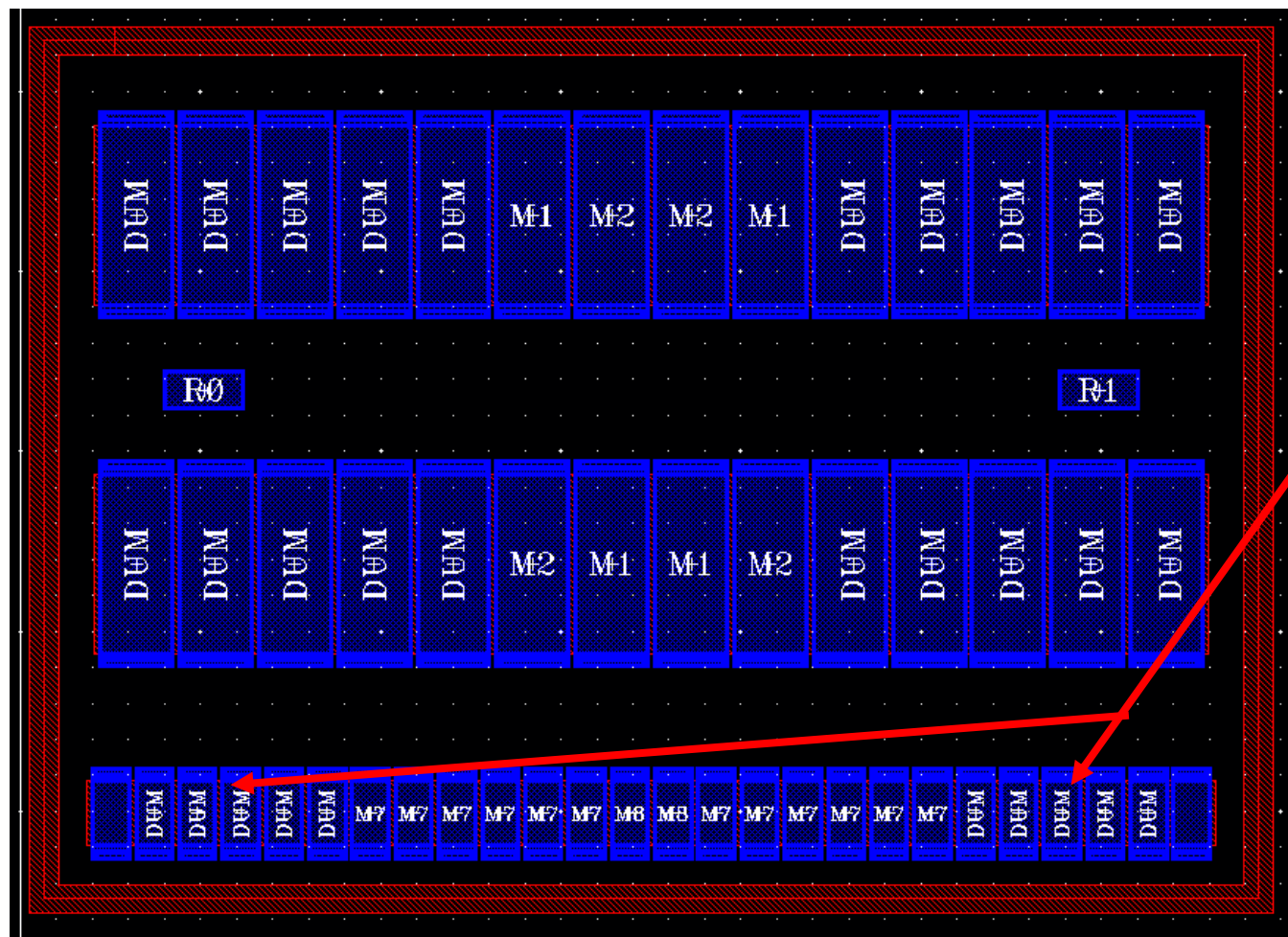
◆ **Comparator版图 (M1、M2、M7、M8、M17、R0、R1布局)** ; 1、M17为Dummy、也为电容;



5.1、Comparator



◆ Comparator版图 (M1、M2、M7、M8、M17、R0、R1布局)¹; M17为Dummy、也为电容;

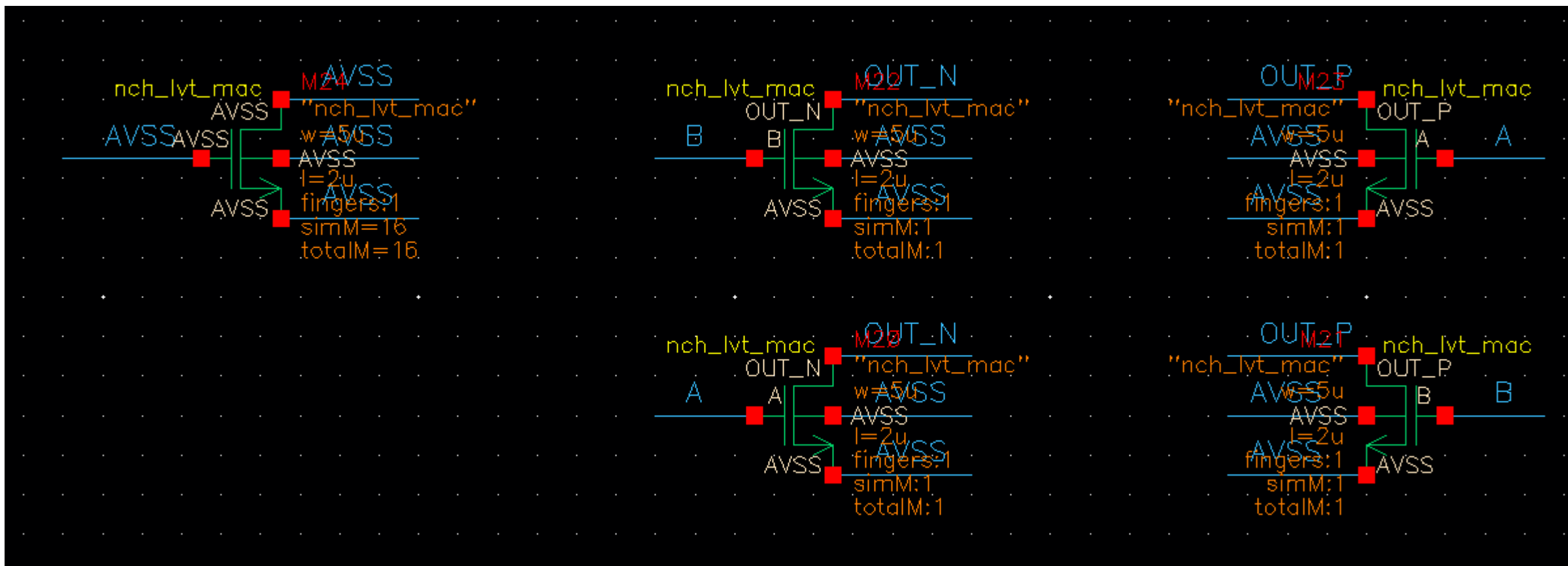


- 1、如左图所示, M1、M2为ABBA二维共质心版图, M8分布在M7中间, 呈对称分布, 并在左右两边加入Dummy MOS;
- 2、电阻呈对称分布, 加入电阻的目的在后面进行说明;



5.1、Comparator

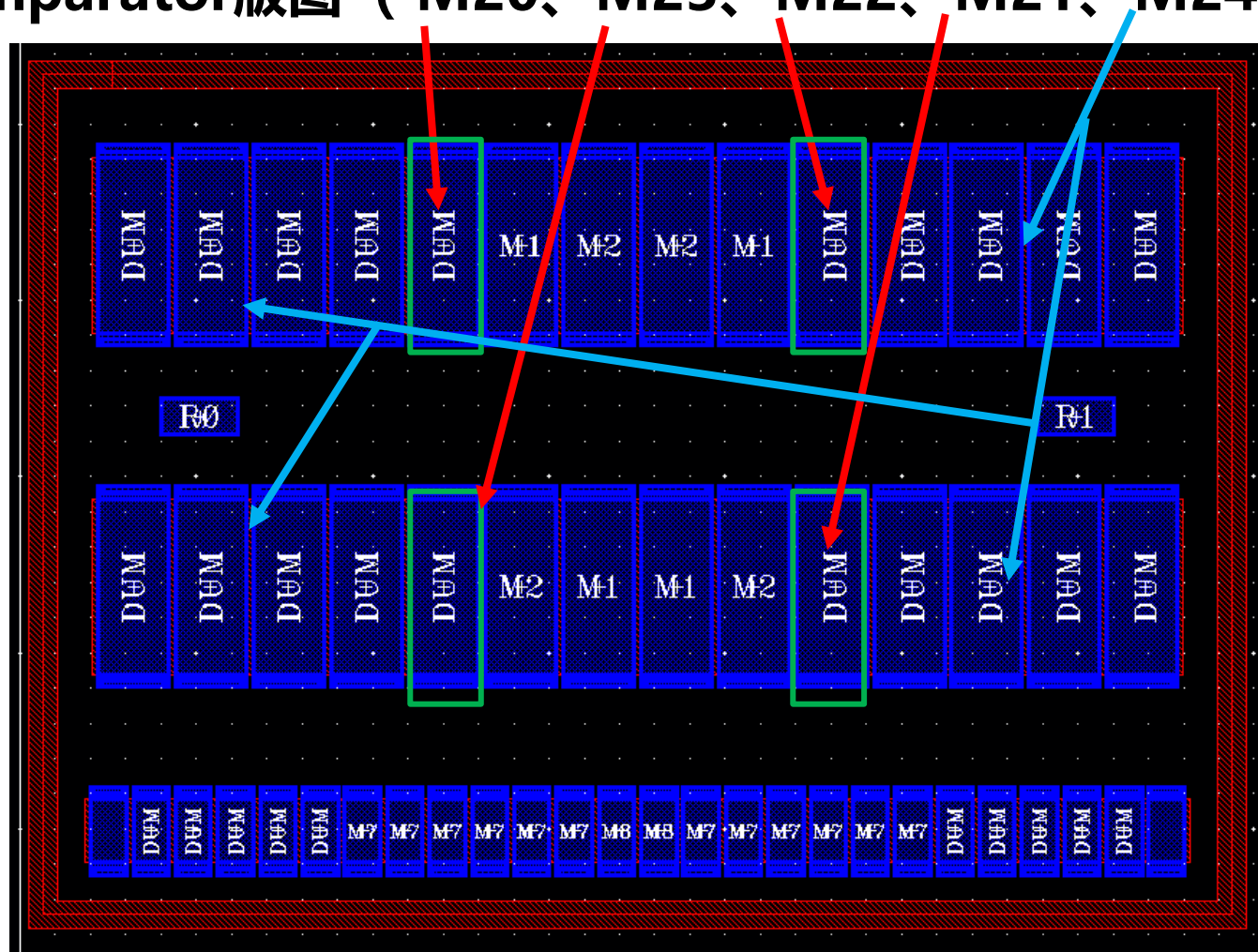
◆ Comparator版图 (M20、M23、M22、M21、M24布局, 均为Dummy MOS) ;





5.1、Comparator

◆ Comparator版图 (M20、M23、M22、M21、M24布局, 均为Dummy MOS) ;

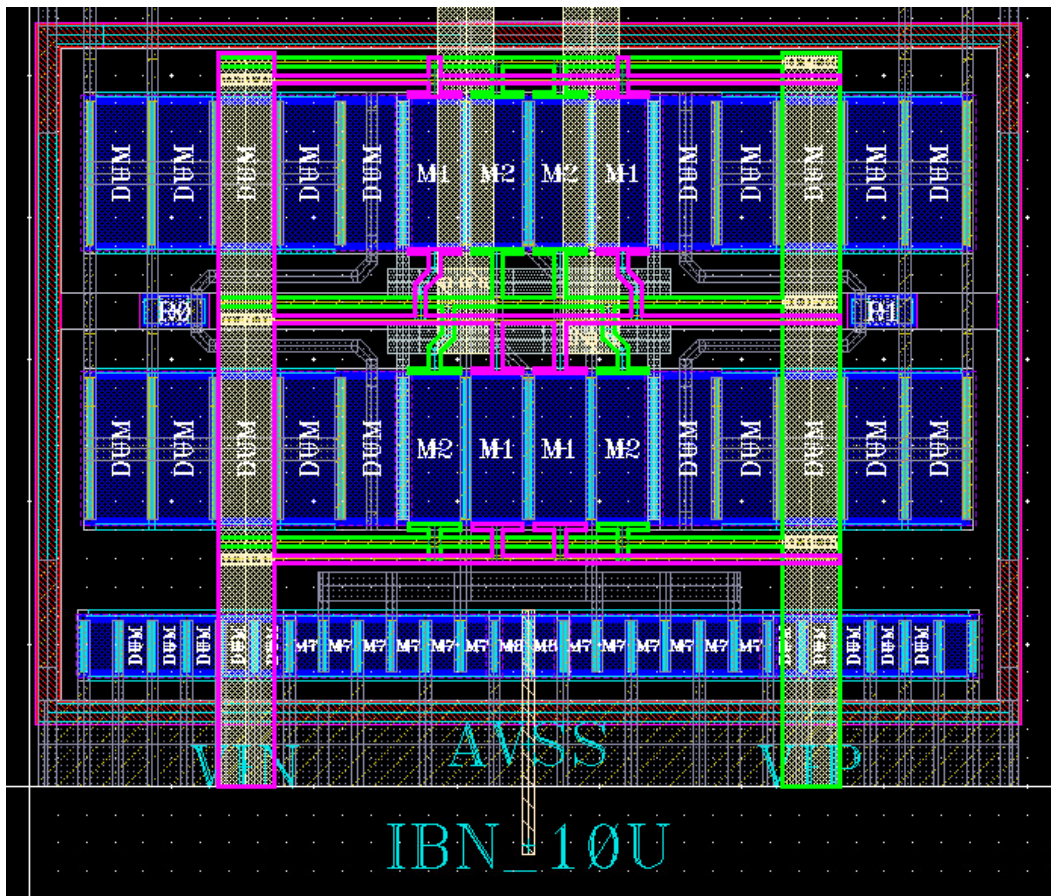


1、加入电阻的是因为M20、M23、M22、M21作为Dummy MOS, 一端Source与M1、M2共用, 其余三端接地时会有栅极直接接地的ERC问题, 故VSS过电阻后再接入Dummy MOS的栅极, 起保护作用;

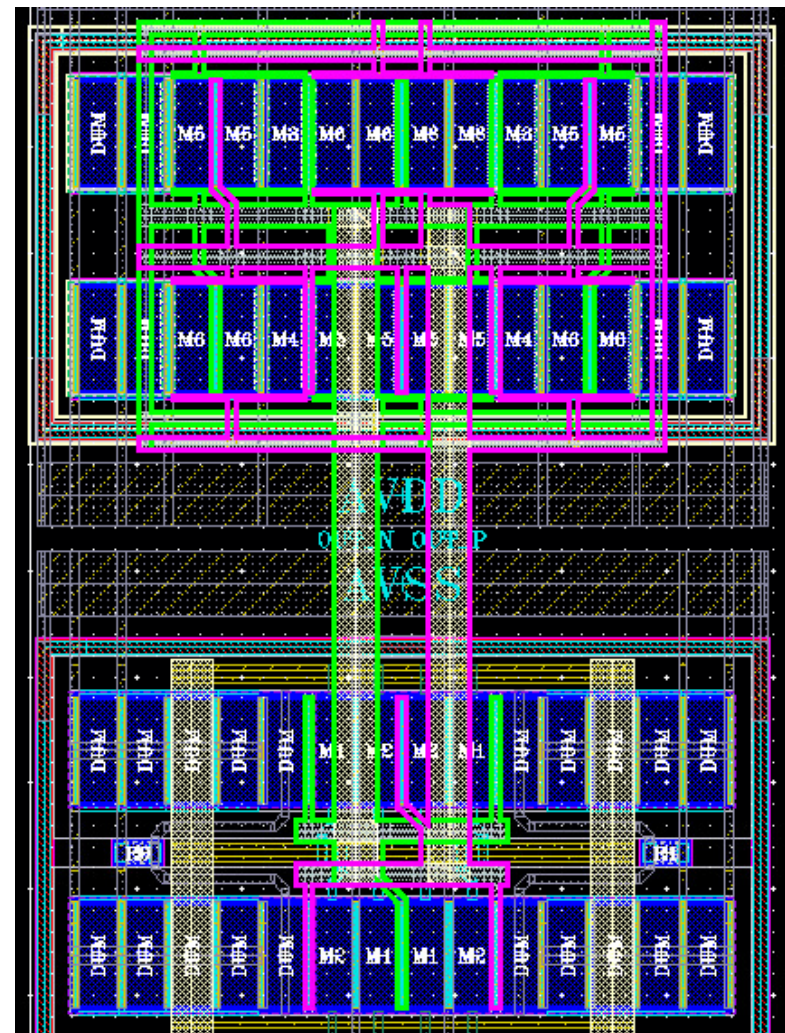


5.1、Comparator

◆ Comparator版图关键信号走线;



1、Comparator layout VIN、VIP;

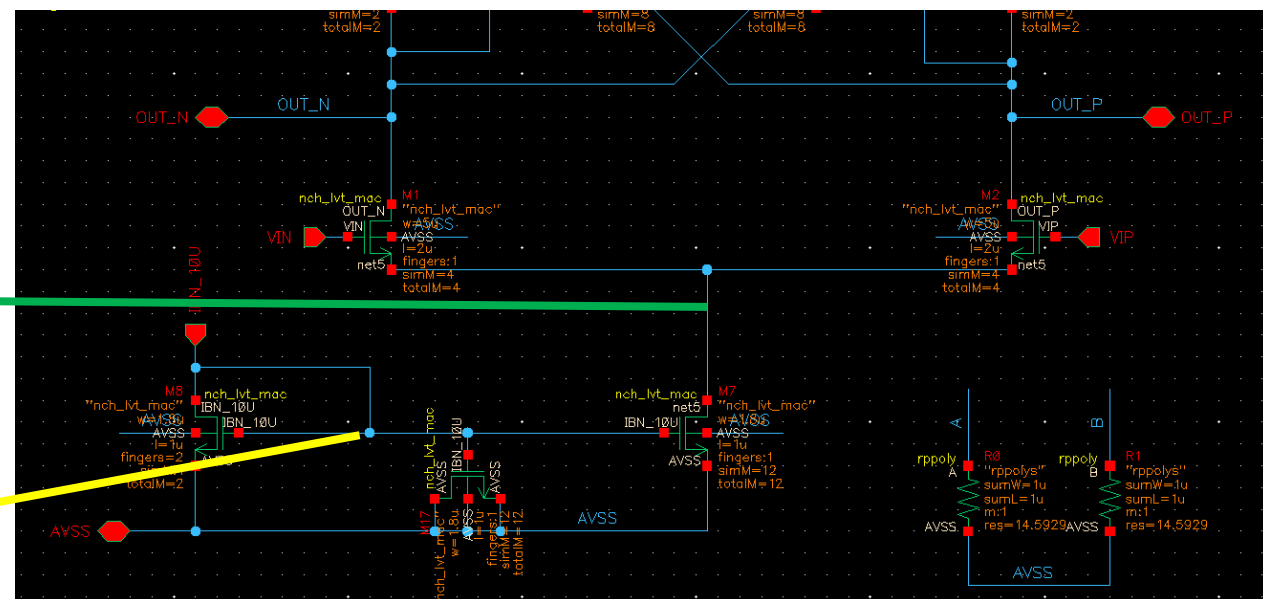
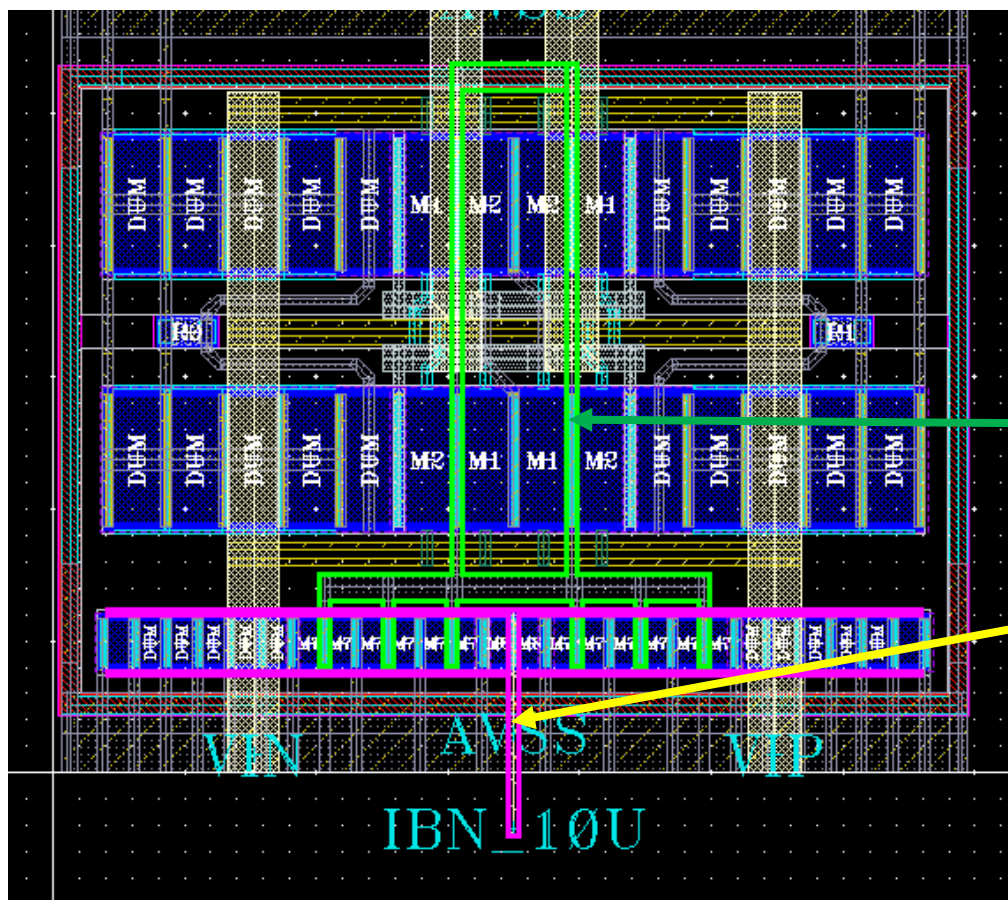


1、Comparator layout OUT_N、OUT_P;



5.1、Comparator

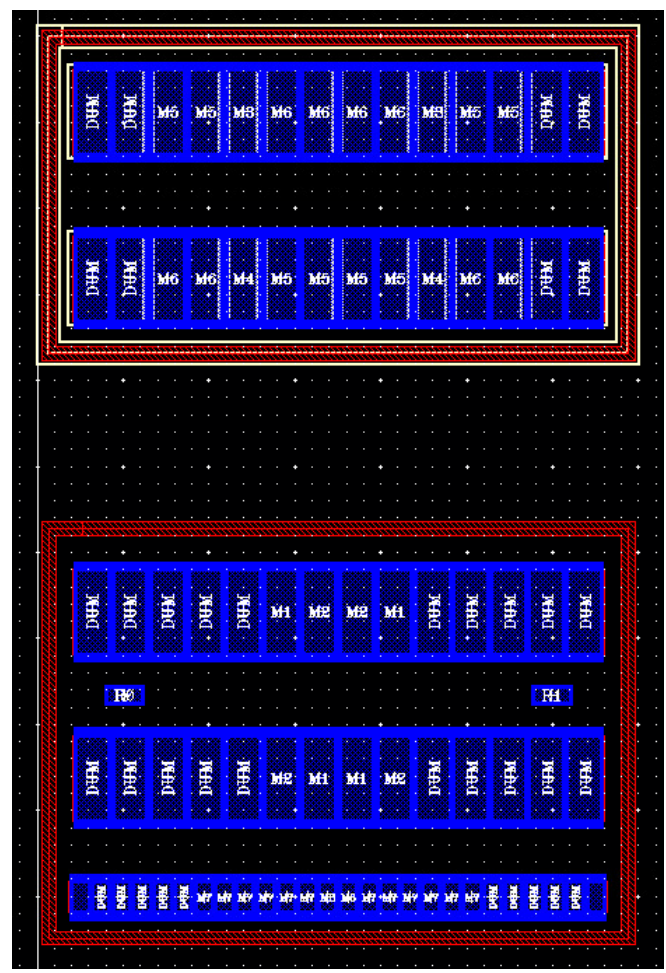
◆ Comparator版图关键信号走线;



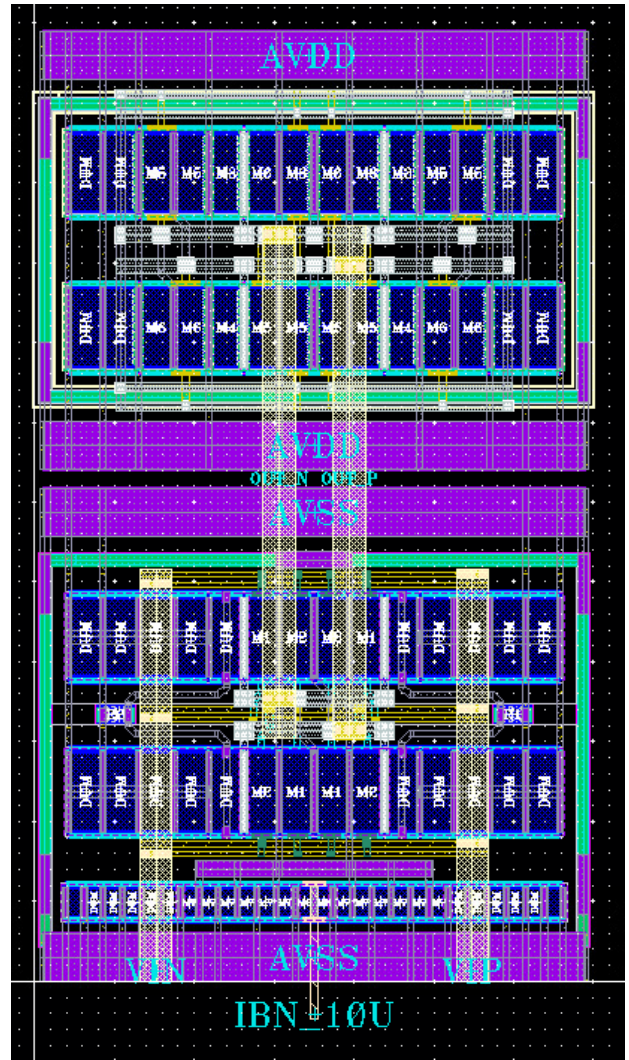
5.1、Comparator

◆ Comparator版图整体布局与最终版图;

1、Comparator 整体布局;



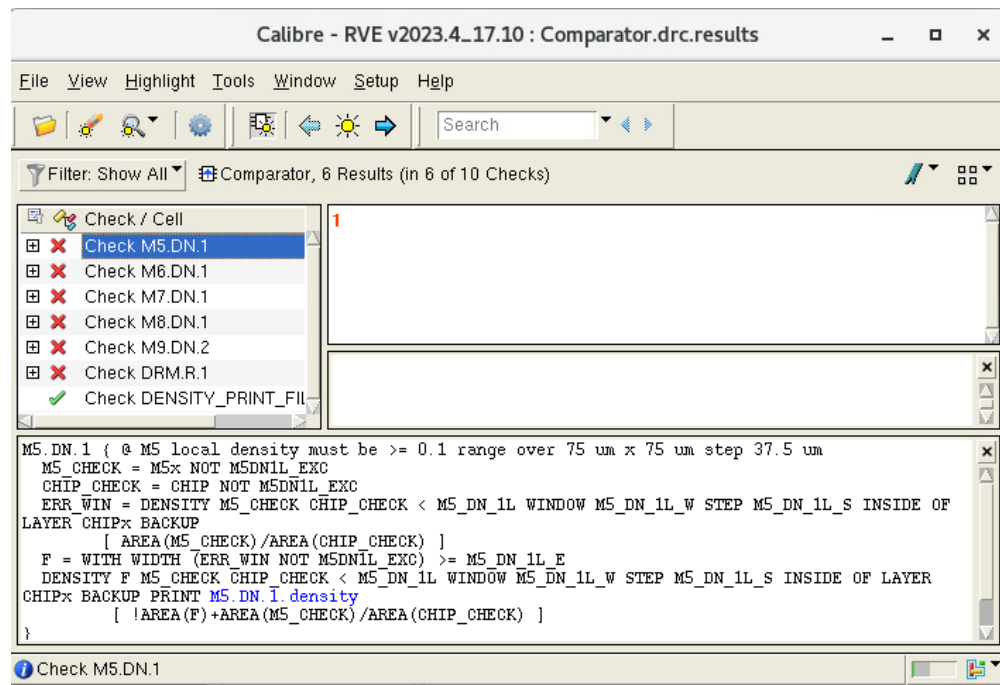
2、Comparator 最终版图;



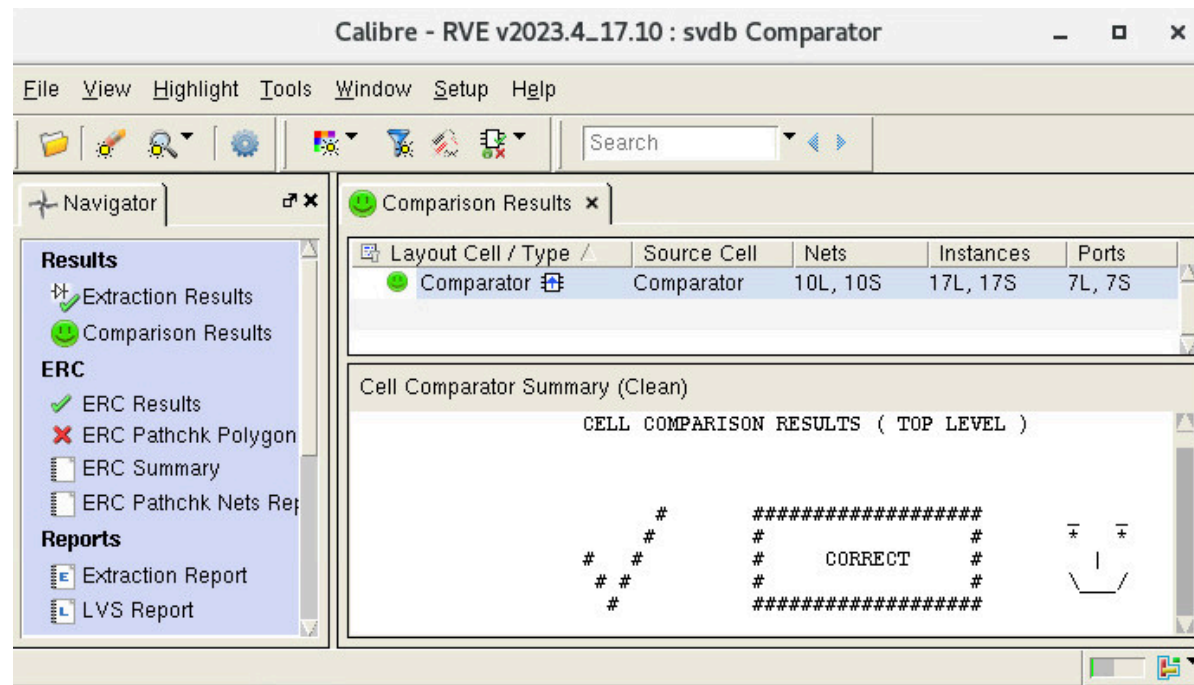


5.1、Comparator

◆ Comparator版图;



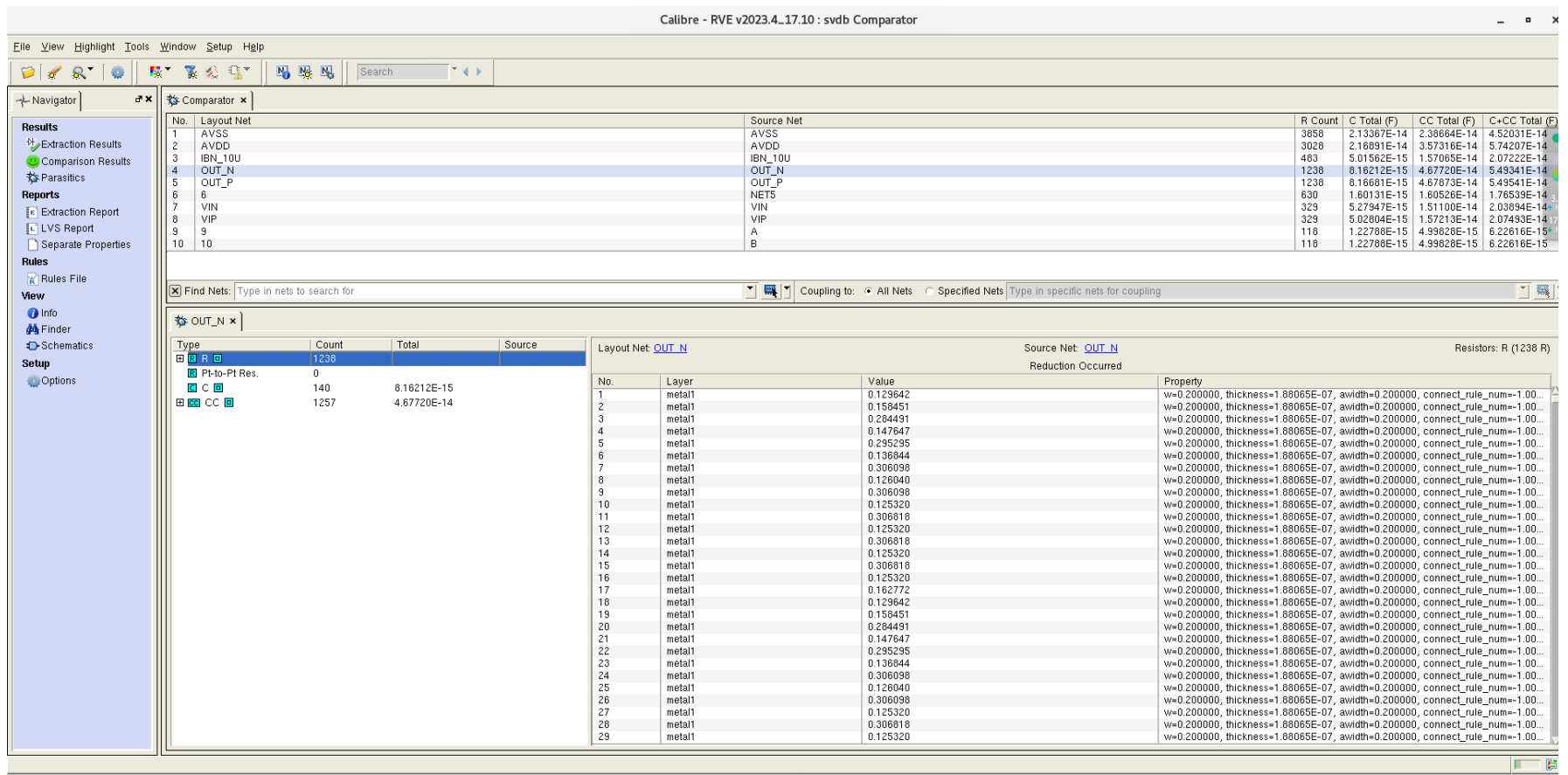
1、Comparator layout **DRC** Results
只有密度过小的DRC，模块级别
可以暂时不用修正



2、Comparator layout **LVS** Results

5.1、Comparator

◆ Comparator版图;

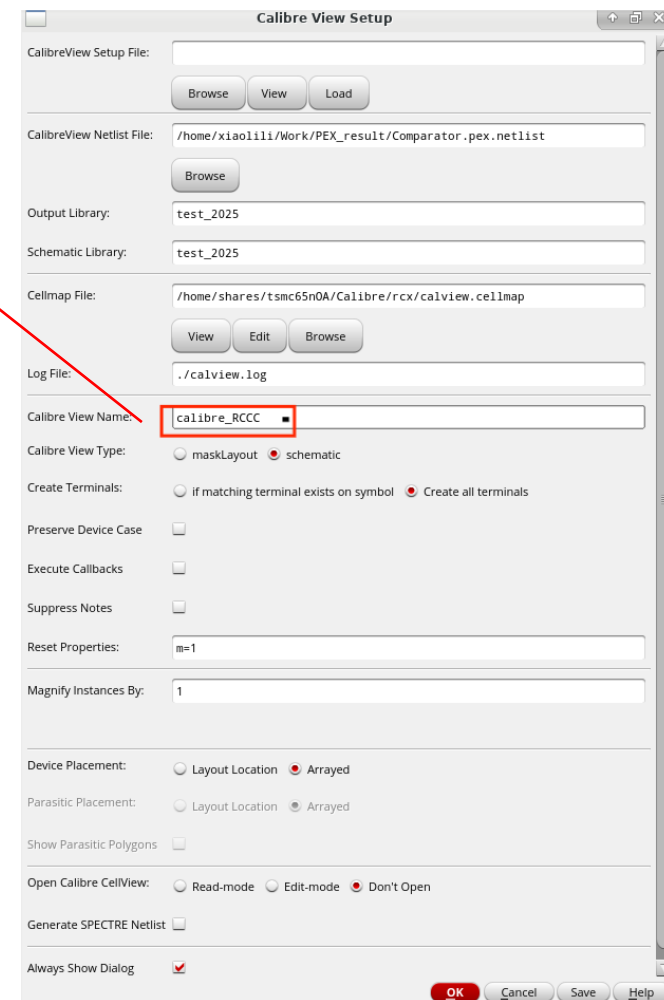
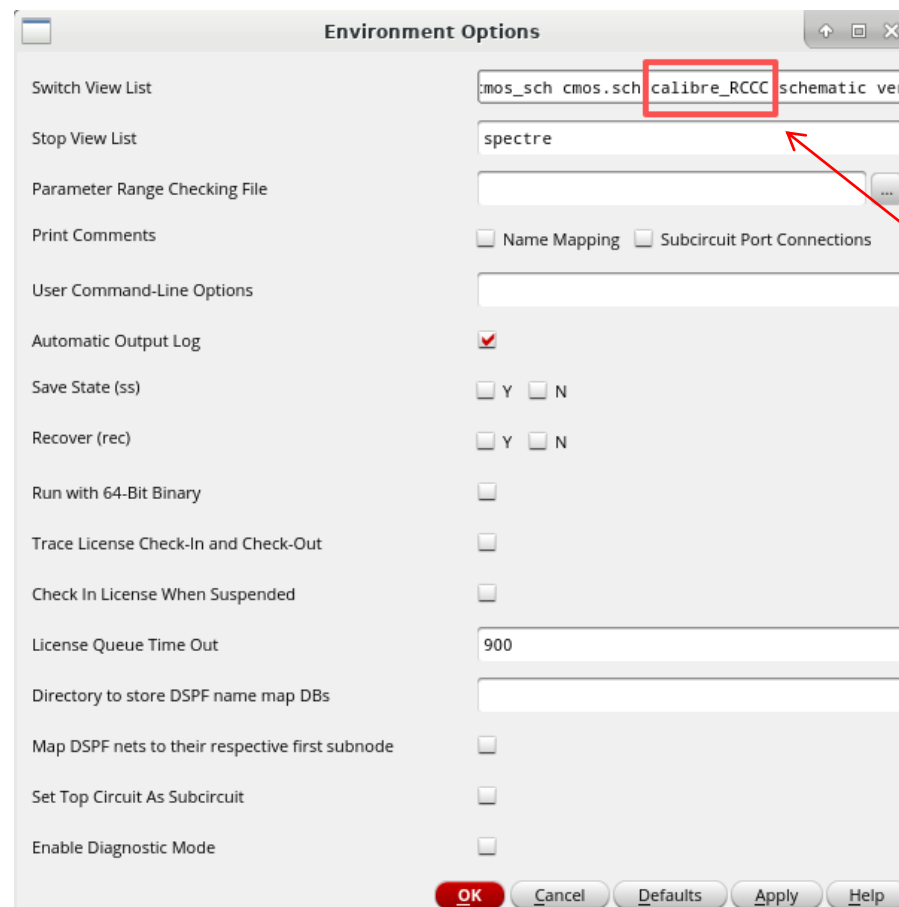
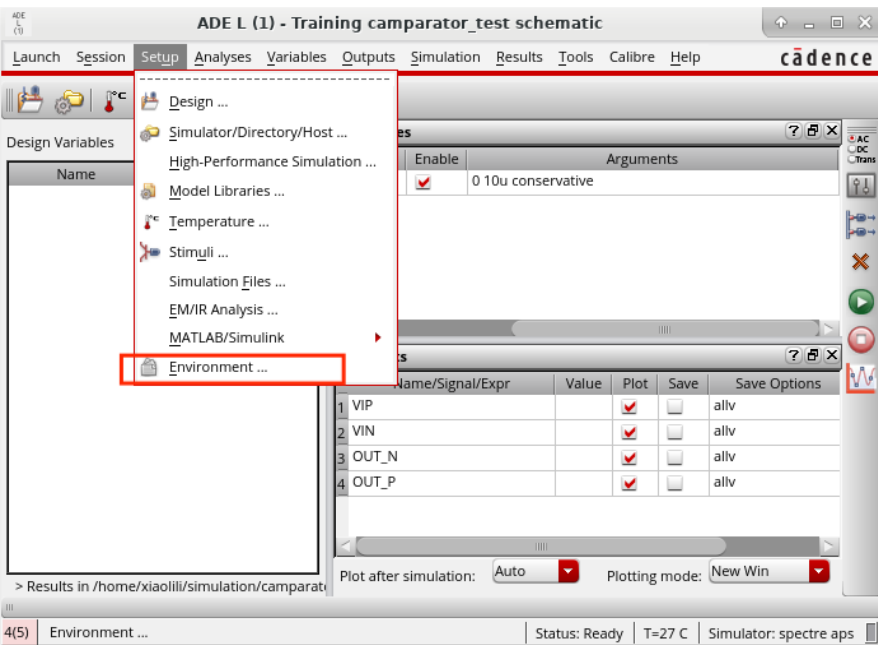


1、Comparator layout PEX Results

5.1、Comparator瞬态仿真——后仿



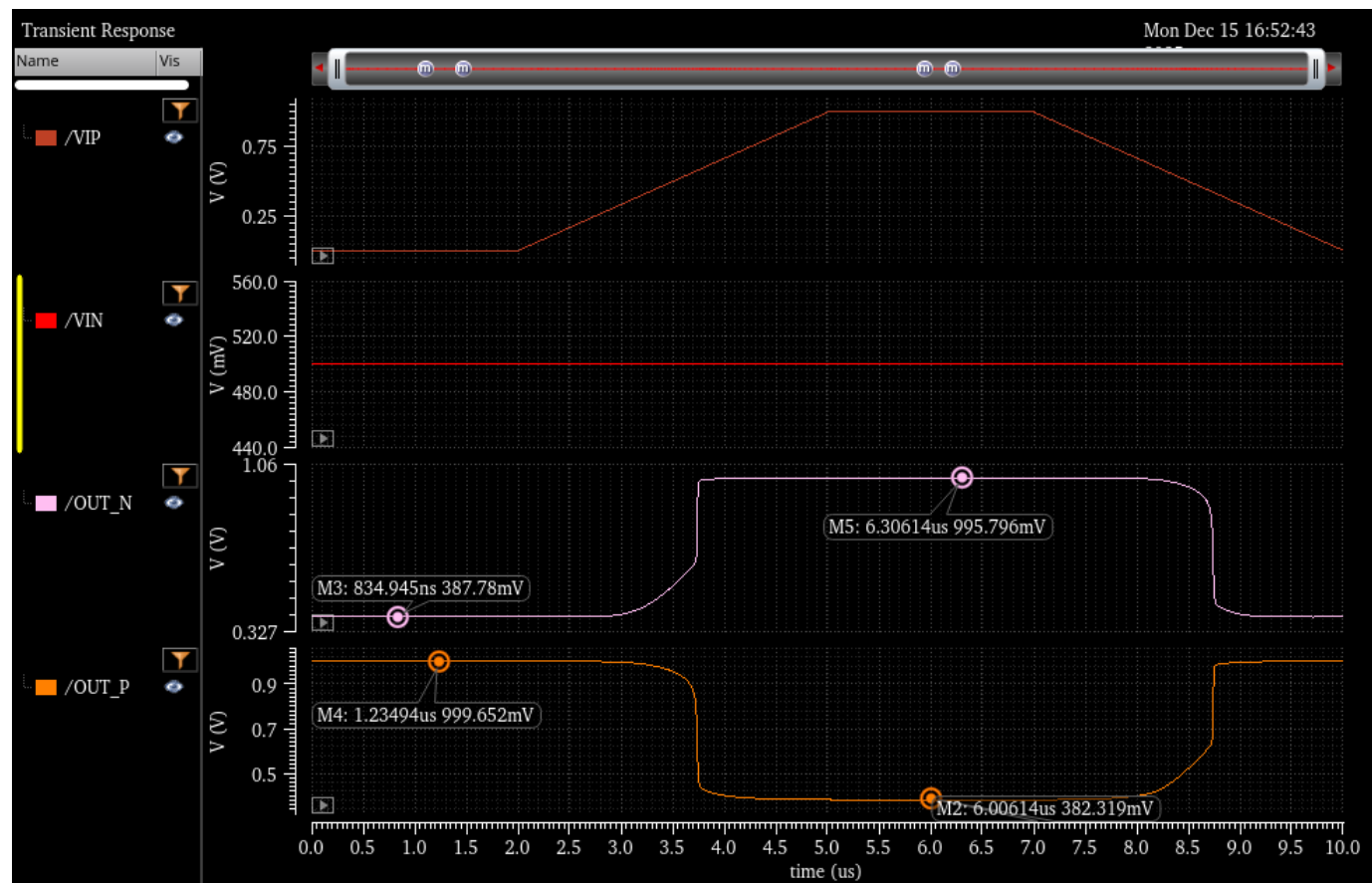
e) 对所设计的版图进行寄生提取后进行后仿真，打印输入VIP，VIN，输出OUTN/OUTP节点的波形，并解释；



5.1、Comparator瞬态仿真——后仿



e) 对所设计的版图进行寄生提取后进行后仿真，打印输入VIP，VIN，输出OUTN/OUTP节点的波形，并解释；



与前仿基本一致，输出电压由于受寄生电阻和寄生电容的影响，稍有差异。



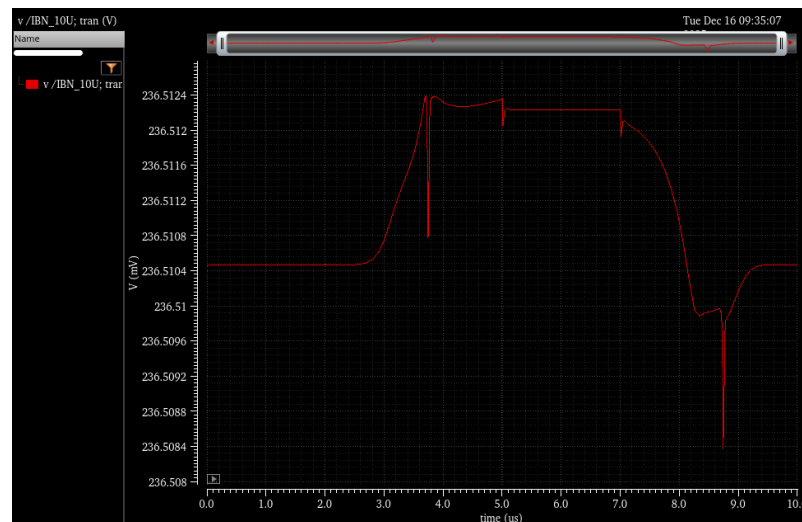
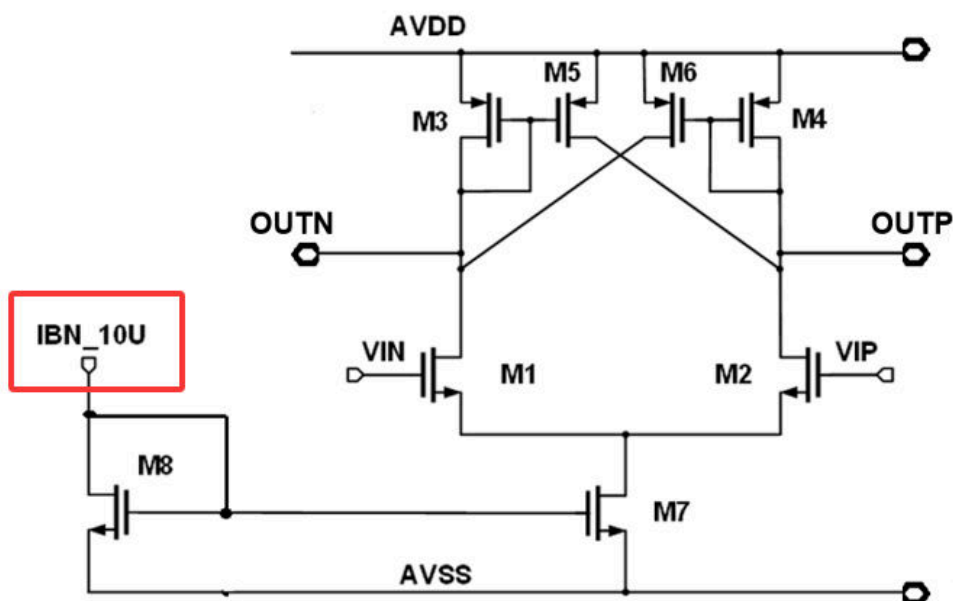
5.1、Comparator瞬态仿真——前后仿对比

		前仿	后仿
OUT_P	上升时间/ns	477.7	472.9
	下降时间/ns	193.4	190.1
	峰峰值/V	0.609	0.618
OUT_N	上升时间/ns	466.7	457.1
	下降时间/ns	187.6	189.3
	峰峰值/V	0.596	0.608
DC功耗/uW		56.19	59.46

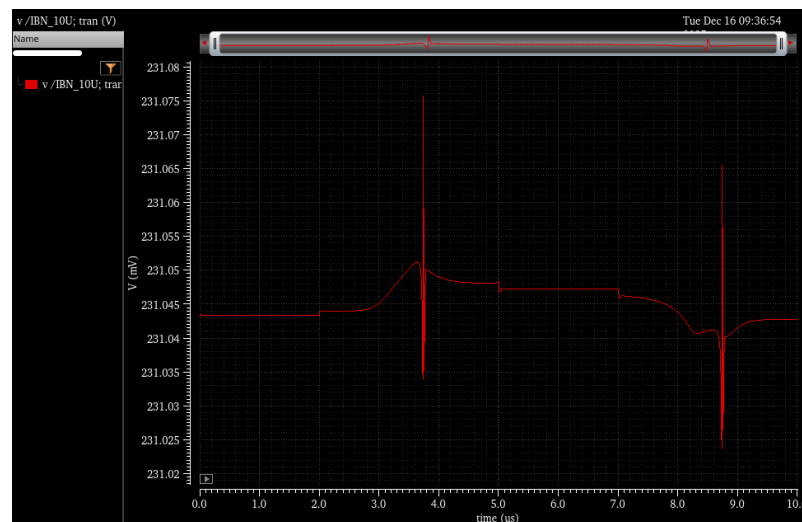
5.1、Comparator瞬态仿真——前后仿对比



◆ IBN_10U处的电压



前仿：电压值约为
236.5mV

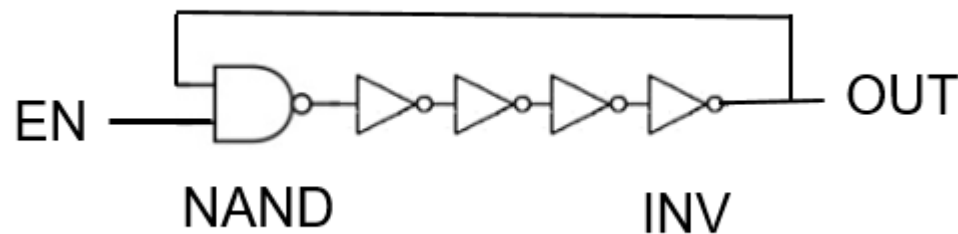


后仿：电压值约为
231mV

5.2、Oscillator



◆ 定制数字的部分版图，设计一个带使能信号的5级基于反相器的环形振荡器电路仿真版图。



5.2、Oscillator

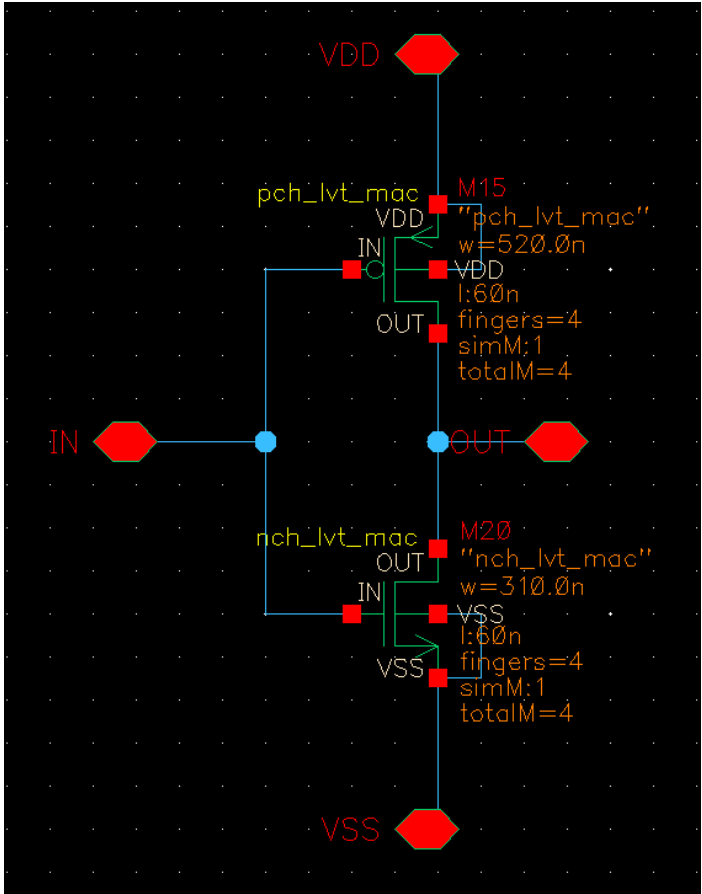


◆ 要求;

- 1) 当使能信号为1时, 环形振荡器起振, 否则不起振;
- 2) 在Cadence环境下完成芯片的电路图, 并建立其Symbol;
- 3) 完成PVT (tt/ff/ss, $V=0.9/1/1/1V$ $T=-20/25/85^{\circ}$ 总计27个case下振荡器频率);
- 4) 在ADE环境下完成瞬态仿真, 打印输出OUT节点的波形;
- 5) TT/1V/27°情况下, 进行蒙特卡洛仿真, 并测试其中心频率的 μ 和 σ
- 6) 完成该芯片的版图设计, 并进行DRC和LVS检查;
- 7) 对所设计的版图进行寄生提取后进行后仿真, 并获得计算LPE震荡频率;

5.2、Oscillator

◆ INV_4X电路搭建;



1、INV_4X schematic;

3、pch_lvt_mac参数;

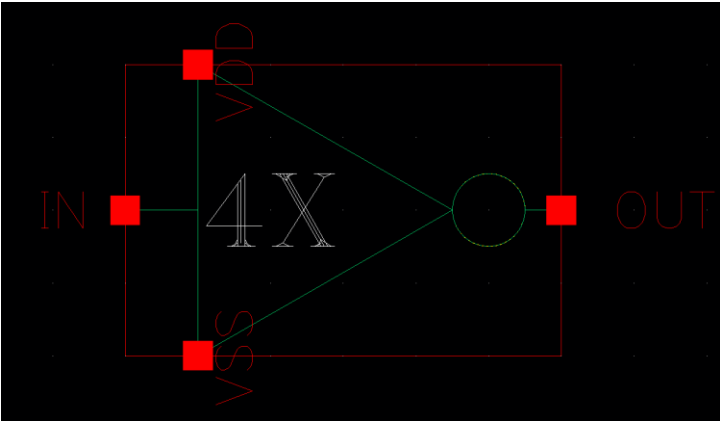
Library Name
tsmcN65

Cell Name
pch_lvt_mac

View Name
symbol1

Instance Name
M15

CDF Parameter	Value	Display
Model name	pch_lvt_mac	off
description	V LOW VT PMOS transistor	off
I (M)	60n M	off
w (M)	520.0n M	off
total_width(M)	2.08u M	off
Number of Fingers	4	off
Multiplier	1	off
total_m	4	off
Hard_constraint	☒	off
With Mismatch Effect	☒	off
Mismatch_Sigma	1	off
Calc Diff Params	☒	off



2、INV_4X symbol;

4、nch_lvt_mac参数;

Library Name
tsmcN65

Cell Name
nch_lvt_mac

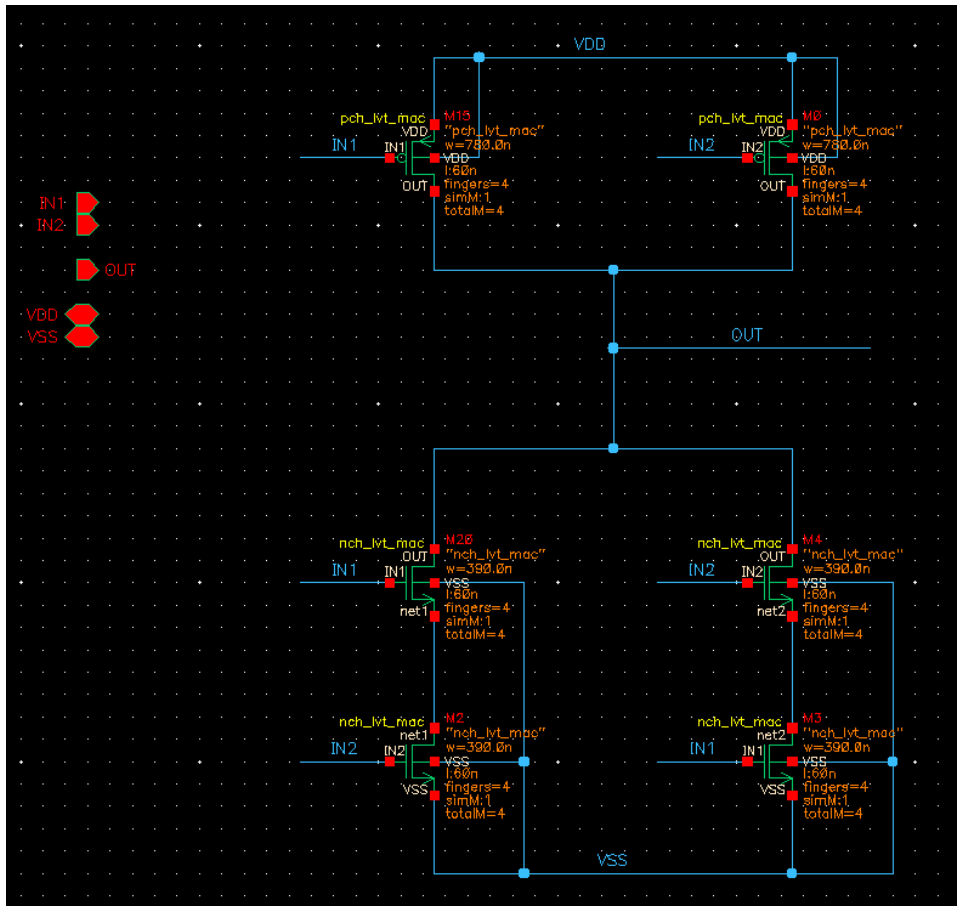
View Name
symbol1

Instance Name
M20

CDF Parameter	Value	Display
Model name	nch_lvt_mac	off
description	LOW VT NMOS transistor	off
I (M)	60n M	off
w (M)	310.0n M	off
total_width(M)	1.24u M	off
Number of Fingers	4	off
Multiplier	1	off
total_m	4	off
Hard_constraint	☒	off
With Mismatch Effect	☒	off
Mismatch_Sigma	1	off

5.2、Oscillator

◆ NAND_4X电路搭建;

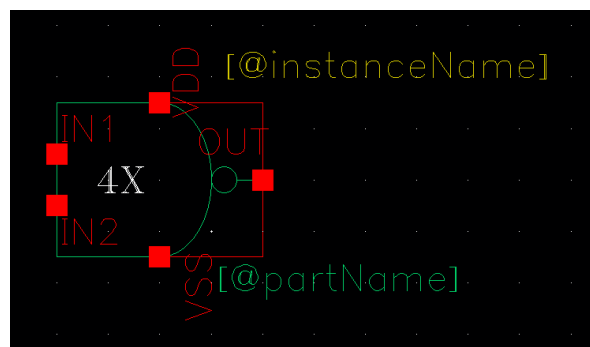


1、NAND_4X schematic;

3、pch_lvt_mac参数;

Library Name	tsmcN65
Cell Name	pch_lvt_mac
View Name	symbol
Instance Name	M15

CDF Parameter	Value	Display
Model name	pch_lvt_mac	off
description	V LOW VT PMOS transistor	off
I (M)	60n M	off
w (M)	780.0n M	off
total_width(M)	3.12u M	off
Number of Fingers	4	off
Multiplier	1	off
total_m	4	off
Hard_constrain	☒	off
With Mismatch Effect	☒	off
Mismatch_Sigma	1	off



2、NAND_4X symbol;

4、nch_lvt_mac参数;

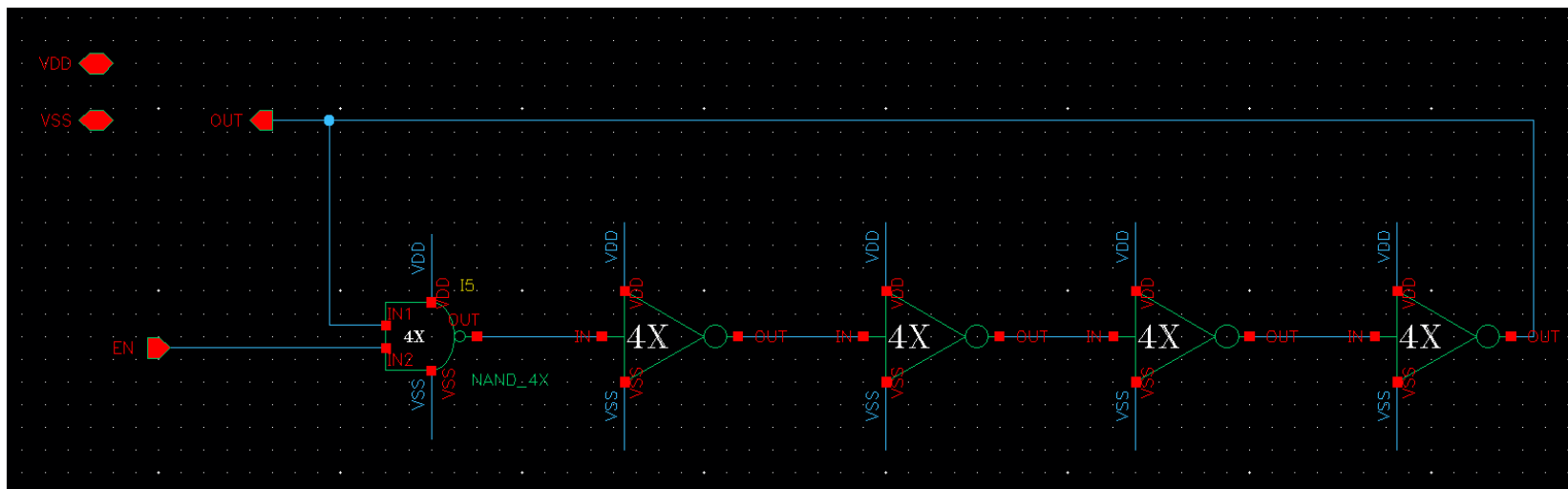
Library Name	tsmcN65
Cell Name	nch_lvt_mac
View Name	symbol
Instance Name	M2

CDF Parameter	Value	Display
Model name	nch_lvt_mac	off
description	LOW VT NMOS transistor	off
I (M)	60n M	off
w (M)	390.0n M	off
total_width(M)	1.56u M	off
Number of Fingers	4	off
Multiplier	1	off
total_m	4	off
Hard_constrain	☒	off
With Mismatch Effect	☒	off
Mismatch_Sigma	1	off

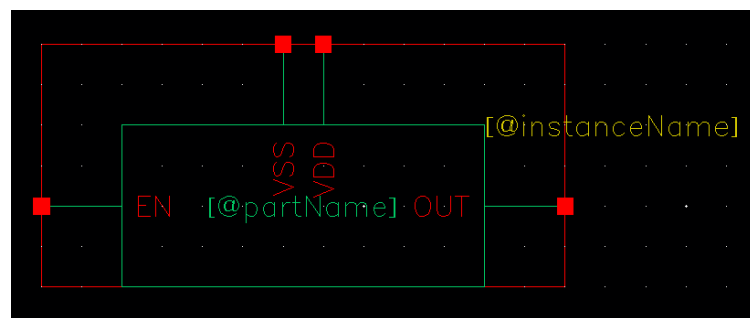
5.2、Oscillator



◆ Oscillator电路搭建;



1、Oscillator schematic;

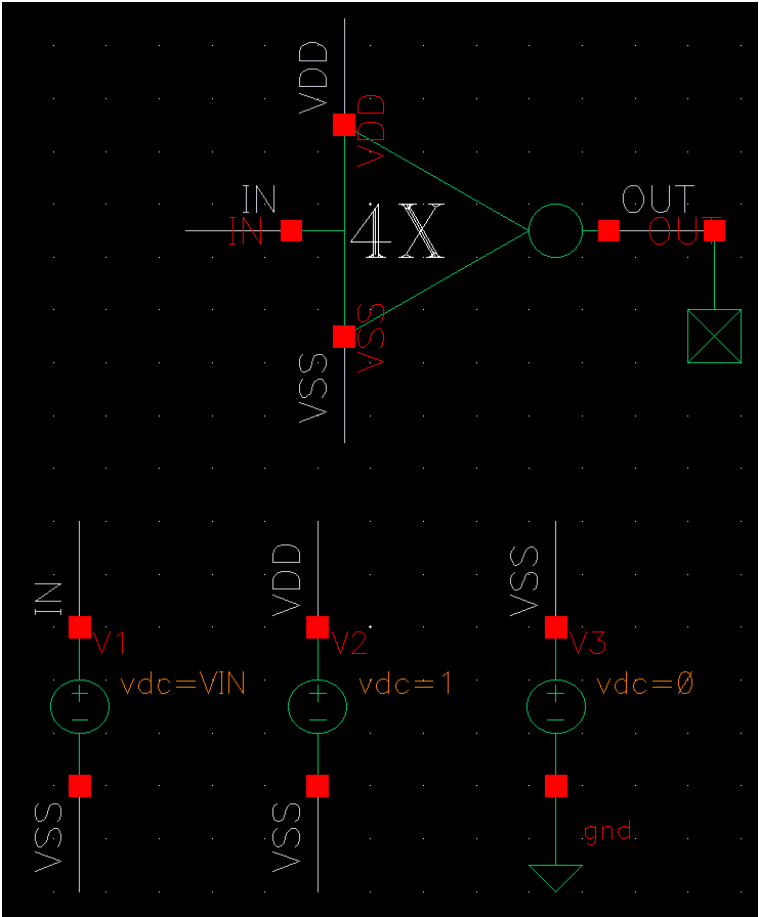


2、Oscillator symbol;

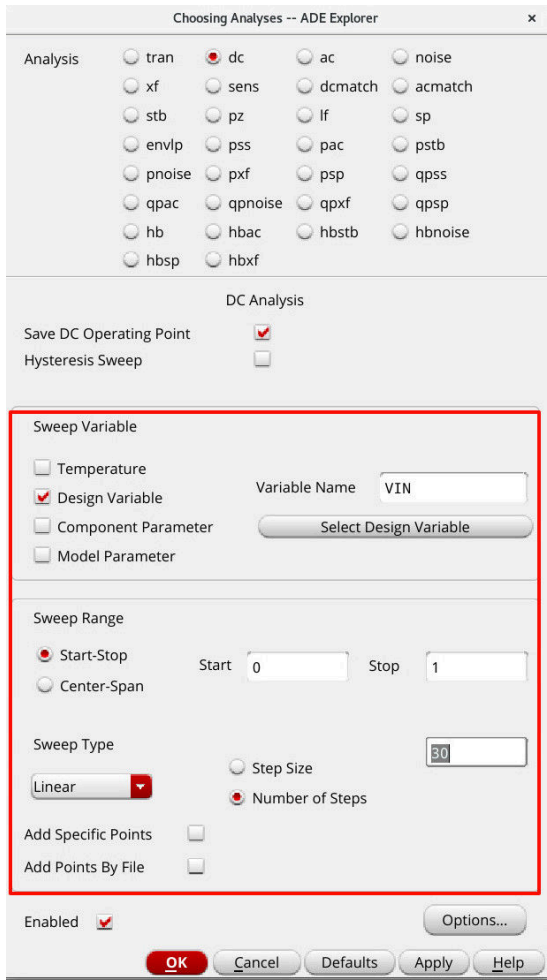


5.2、Oscillator INV_X4传输曲线仿真——前仿

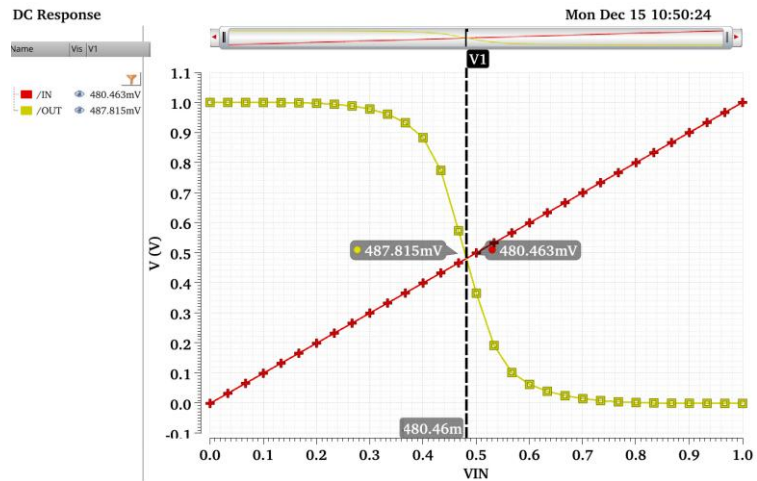
测试电路图



仿真设置



DC 扫描反相器输入 V_{in} ，仿真 V_{in}/V_{out} 的传输曲线。

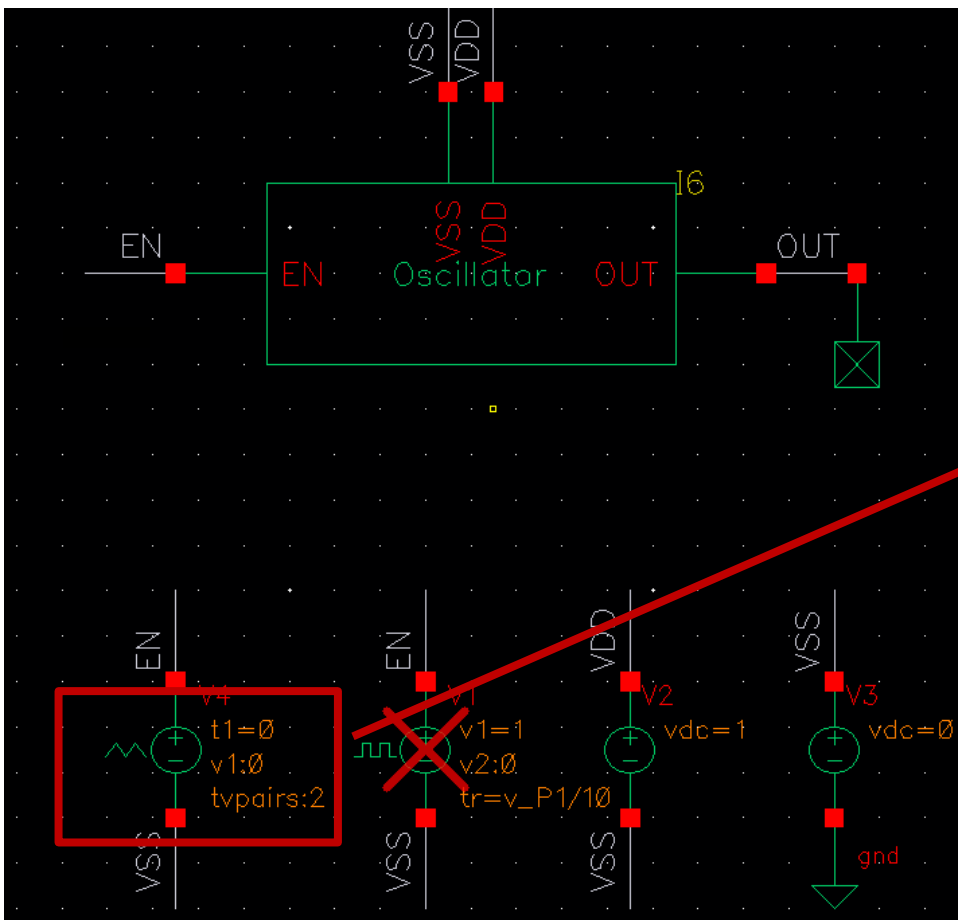


接近(500, 500)的中点，逻辑0 1区域对称，证明P和N的尺寸设置合理。

5.2、Oscillator输出波形仿真——前仿



测试电路图



使用 vpwl 器件，可以设置不同时间的电压，用来产生0到1的上升沿，使振荡器震荡。

Property	Value	Display
Library Name	analogLib	off
Cell Name	vpwl	off
View Name	symbol	off
Instance Name	V4	off

User Property	Master Value	Local Value	Display
Frequency name for 1/period			
Number of pairs of points	2		
Time 1	0 s		
Voltage 1	0 V		
Time 2	50n s		
Voltage 2	1 V		
Noise file name			
Number of noise/freq pairs	0		
DC voltage	1 V		

共两个点，首先是 $t=0$ 时电压设置为0V。
之后 $t=50\text{ns}$ 时，电压变化为1V。



5.2、Oscillator输出波形仿真——前仿

仿真设置 设置1微秒的瞬态仿真

Name	Value
Filter	Filter
class_test_2025_tb_oscillator_1	
Simulator	spectre
Analyses	
☒ tran	0 1u conservative
Click to add analysis	
Design Variables	
☐ v_P1	0
Click to add variable	
Parameters	
Corners	
Reliability Analyses	
Monte Carlo Sampling	
Checks/Asserts	

Name	Type	Details
	signal	/EN
	signal	/OUT

Choosing Analyses -- ADE Explorer

Analysis	☒ tran	☐ dc	☐ ac	☐ noise
	☐ xf	☐ sens	☐ dcmatch	☐ acmatch
	☐ stb	☐ pz	☐ lf	☐ sp
	☐ envlp	☐ pss	☐ pac	☐ pstb
	☐ pnoise	☐ pxf	☐ psp	☐ qpss
	☐ qpac	☐ qpnoise	☐ qpxf	☐ qpsp
	☐ hb	☐ hbac	☐ hbstb	☐ hbnoise
	☐ hbsp	☐ hbxf		

Transient Analysis

Stop Time 1u

Accuracy Defaults (errpreset)

☒ conservative ☐ moderate ☐ liberal

☐ Transient Noise

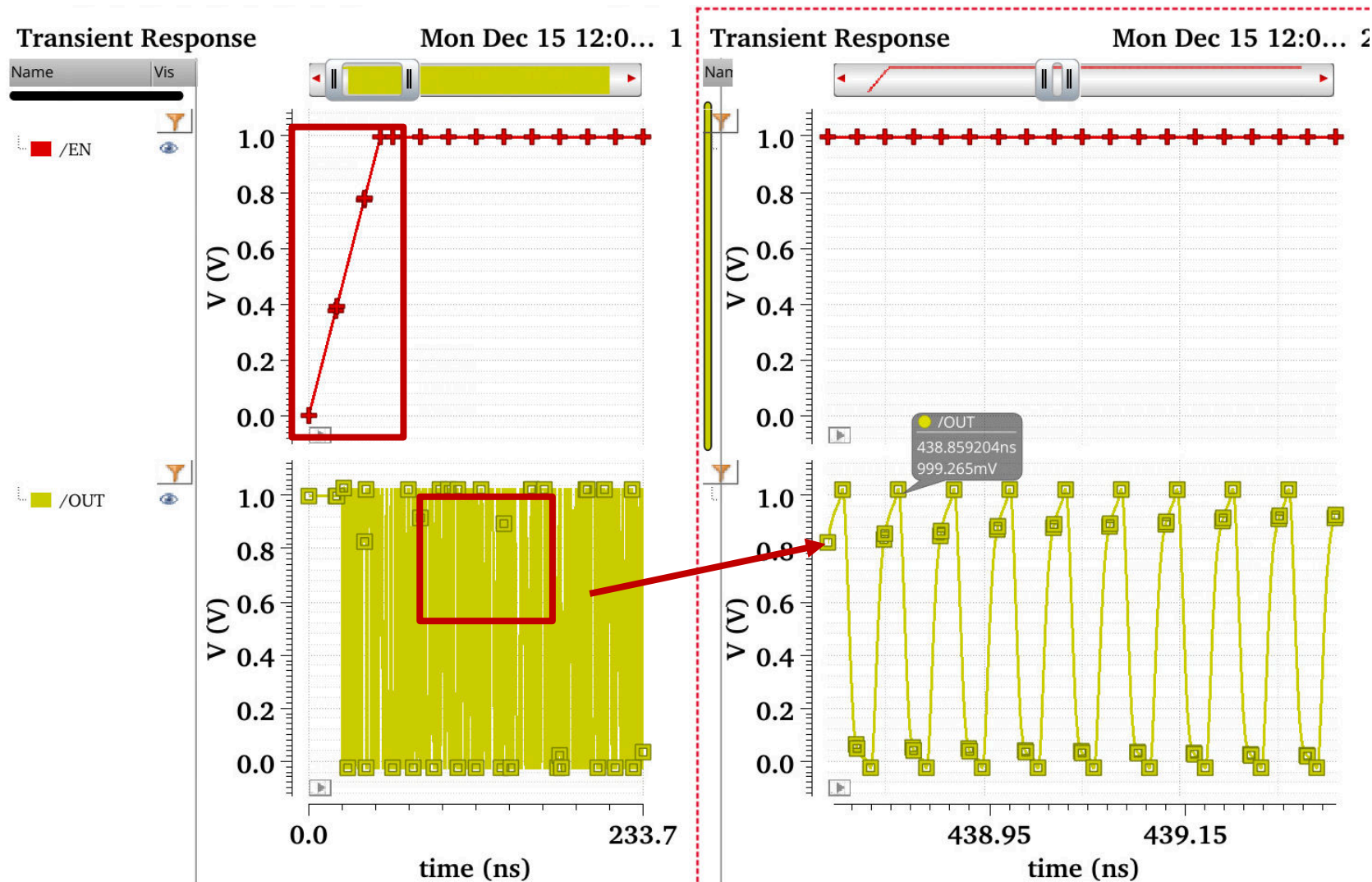
Dynamic Parameter ☐

Enabled ☒

Options...

OK Cancel Defaults Apply Help

5.2、Oscillator输出波形仿真结果——前仿

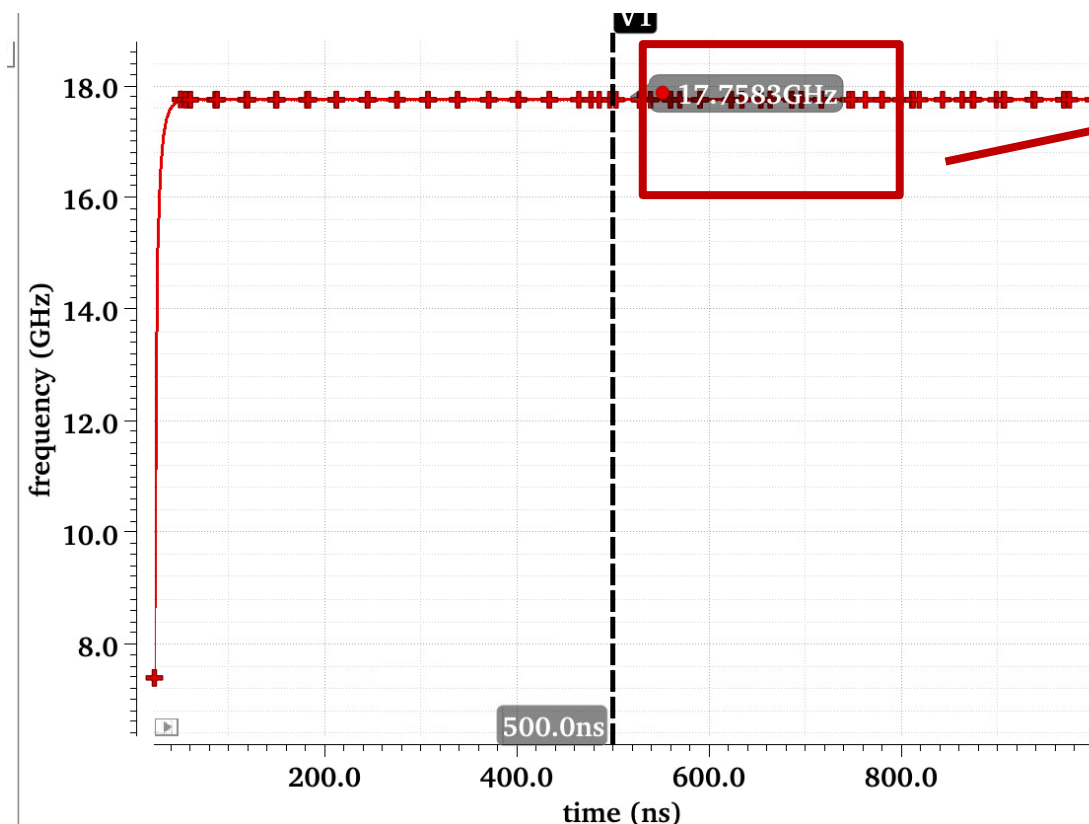


5.2、Oscillator输出波形仿真结果——前仿



使用公式计算不同时间的振荡波形频率，记录500ns时的振荡频率。

w_freq	expr	freq(vtime("tran" "/OUT" nil) "rising" ?xName "time" ?mode "auto" ?threshold 0.0)
	expr	value(freq(vtime("tran" "/OUT" nil) "rising" ?xName "time" ?mode "auto" ?threshold 0.0) 5e-07)

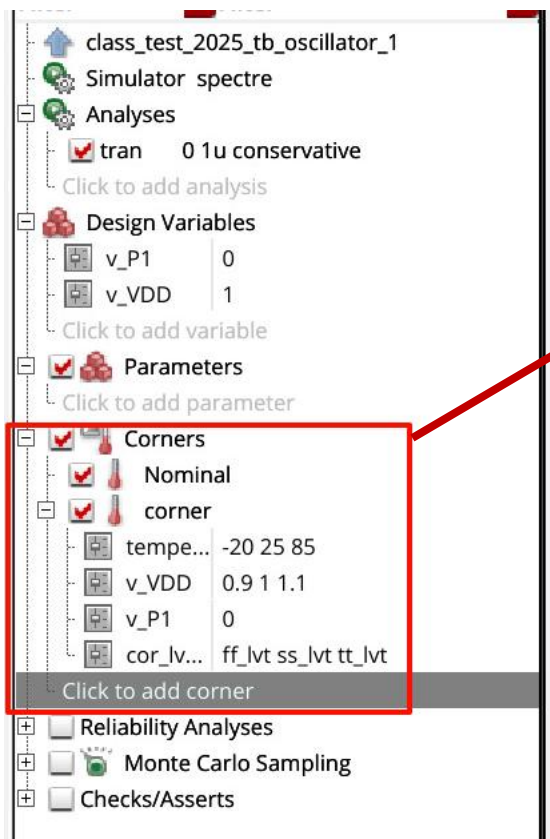


振荡频率为 17.75GHz



5.2、Oscillator Corner仿真——前仿

Test bench 原理图保持不变，修改仿真器设置，增加 corner 仿真设置



Corners	☒ Nominal	☒ corner
Temperature		-20 25 85
Design Variables		
v_VDD		0.9 1 1.1
v_P1		0
Click to add		
Parameters		
Click to add		
Model Files		
cor_lvt.scs		☒ ff_lvt ss_lvt tt_lvt
Click to add		
Model Group(s)		<modelgroup>
Click to add		
Tags		
...mber of Corners	1	27

需要设置 model 文件

设置温度为 -20 25 85 三个温度
电压为 0.9 1.0 1.1V三个电压

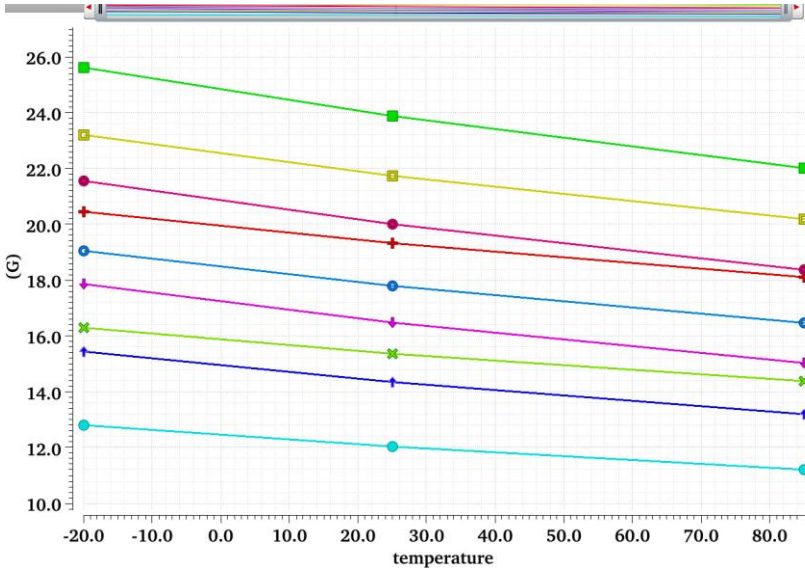
仿真 FF、SS 和 TT 三个工艺角



5.2、Oscillator Corner仿真结果——前仿

Corner 仿真结果

Corner	v_P1	v_VDD	temperature	cor_lvt.scs	Pass/Fail	/EN	w_out	w_freq	..." ?threshold 0.0) 5e-07)
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
nom	0	1	27	<omitted>					17.76G
corner_0	0	900m	-20	ff_lvt					20.46G
corner_1	0	900m	25	ff_lvt					19.33G
corner_2	0	900m	85	ff_lvt					18.12G
corner_3	0	1	-20	ff_lvt					23.21G
corner_4	0	1	25	ff_lvt					21.75G
corner_5	0	1	85	ff_lvt					20.2G
corner_6	0	1.1	-20	ff_lvt					25.63G
corner_7	0	1.1	25	ff_lvt					23.89G
corner_8	0	1.1	85	ff_lvt					22.02G
corner_9	0	900m	-20	ss_lvt					12.81G
corner_10	0	900m	25	ss_lvt					12.05G
corner_11	0	900m	85	ss_lvt					11.22G
corner_12	0	1	-20	ss_lvt					15.45G
corner_13	0	1	25	ss_lvt					14.36G
corner_14	0	1	85	ss_lvt					13.2G
corner_15	0	1.1	-20	ss_lvt					17.87G
corner_16	0	1.1	25	ss_lvt					16.49G
corner_17	0	1.1	85	ss_lvt					15.04G
corner_18	0	900m	-20	tt_lvt					16.3G
corner_19	0	900m	25	tt_lvt					15.37G
corner_20	0	900m	85	tt_lvt					14.39G
corner_21	0	1	-20	tt_lvt					19.06G
corner_22	0	1	25	tt_lvt					17.8G
corner_23	0	1	85	tt_lvt					16.48G
corner_24	0	1.1	-20	tt_lvt					21.56G
corner_25	0	1.1	25	tt_lvt					20.02G
corner_26	0	1.1	85	tt_lvt					18.38G

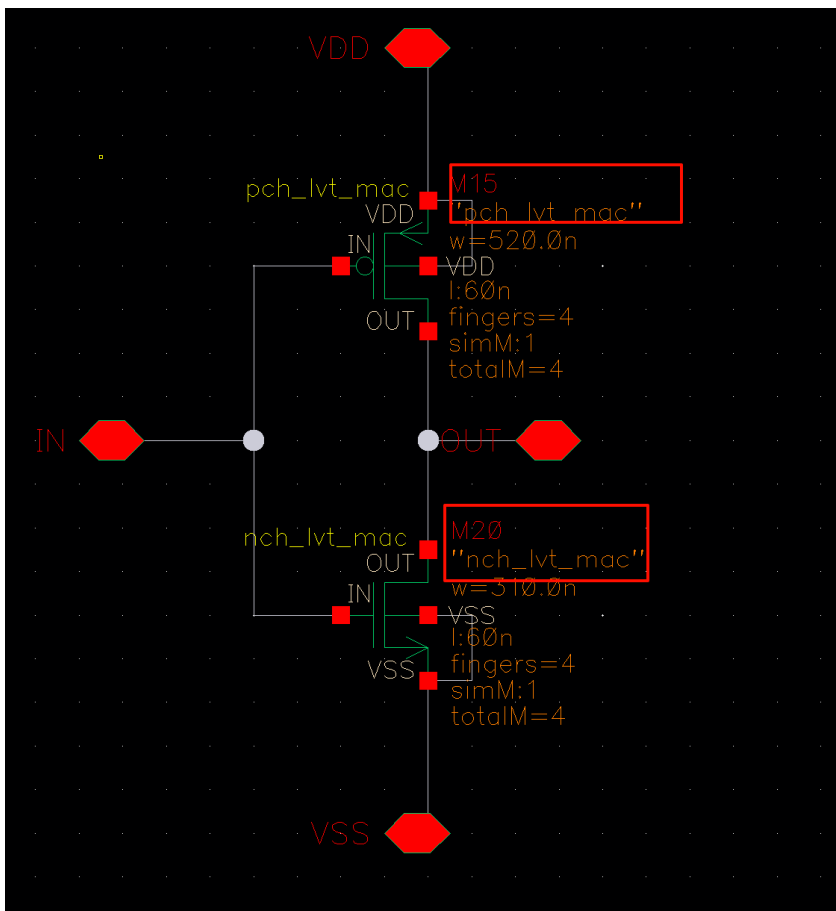


振荡频率随温度的变化

5.2、Oscillator Monte Carlo仿真设置——前仿



将器件更改为携带mismatch参数的器件



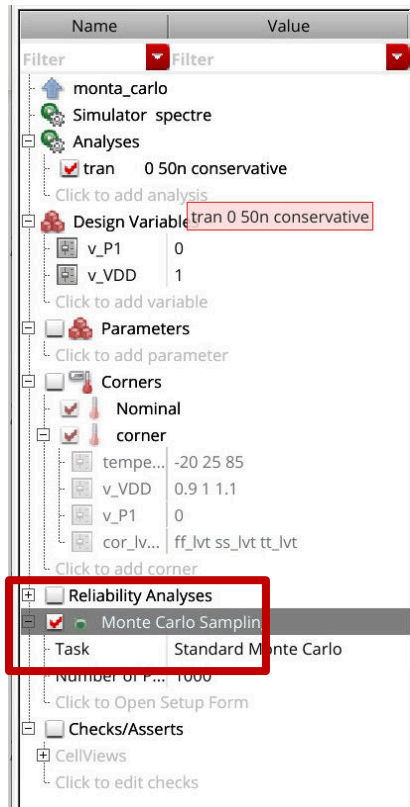
添加工艺统计model 文件, stat_mis_lvt





5.2、Oscillator Monte calro仿真设置——前仿

勾选Monte carlo 仿真



Advanced - Show All Options

Variation: A11

Number of Points: 1000

Points per job: ☒ Group automatically ☐ Max 10

Target Yield: 3 sigma

Method: Standard Monte Carlo

☐ Save Waveforms (Simulation Data)

☐ Save Statistical Parameter Data

Sampling Method: Low-Discrepancy Sequence

Seed: 12345

First Point: 1

Netlist Options:

Specify Instances/Devices (Not Specified)

Specify Mismatch ID (Not Specified)

Specify Design Variables (Not Specified)

设置为1000个点

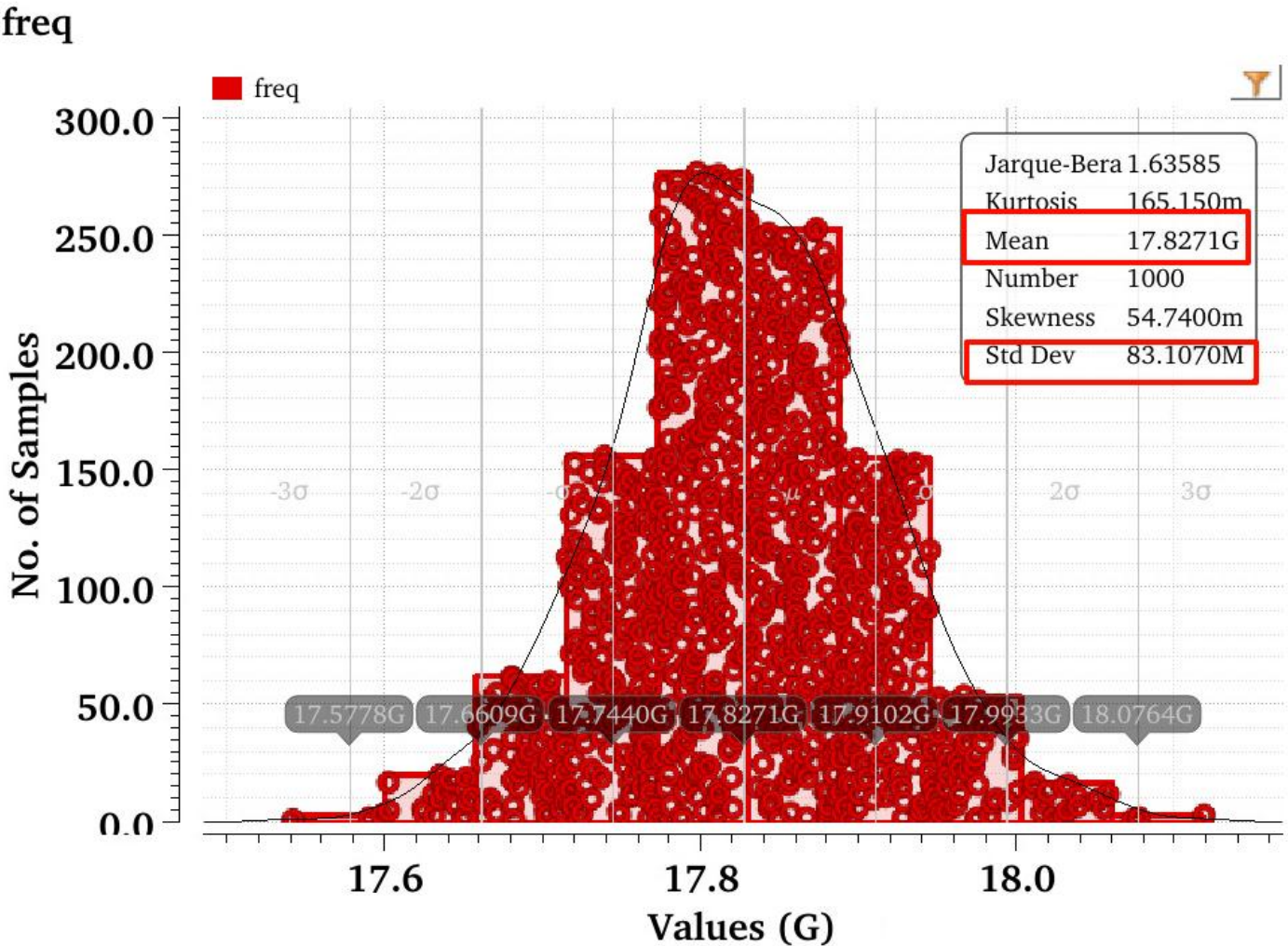


5.2、Oscillator Monte calro仿真结果——前仿

1000次仿真结果

Detail - Transpose			Filter ...			Replace		
1000 rows			1000 rows					
Point	Corner	mc_iteration	Pass/Fail	w_out	freq			
Filter	Filter	Filter	Filter	Filter	Filter			
1	nom	1			17.79G			
2	nom	2			17.68G			
3	nom	3			17.85G			
4	nom	4			17.87G			
5	nom	5			18.03G			
6	nom	6			17.88G			
7	nom	7			17.77G			
8	nom	8			17.86G			
9	nom	9			17.79G			
10	nom	10			17.88G			
11	nom	11			17.81G			
12	nom	12			17.89G			
13	nom	13			17.84G			
14	nom	14			17.72G			
15	nom	15			17.8G			
16	nom	16			17.93G			
17	nom	17			17.7G			
18	nom	18			17.77G			
19	nom	19			17.88G			
20	nom	20			17.84G			
21	nom	21			17.82G			
22	nom	22			17.84G			
23	nom	23			17.97G			

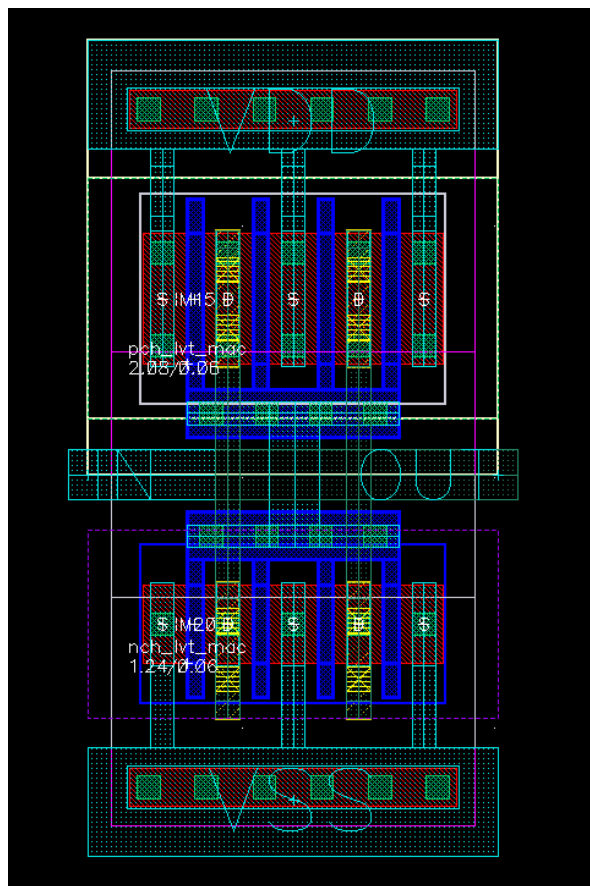
1000次结果的分布直方图



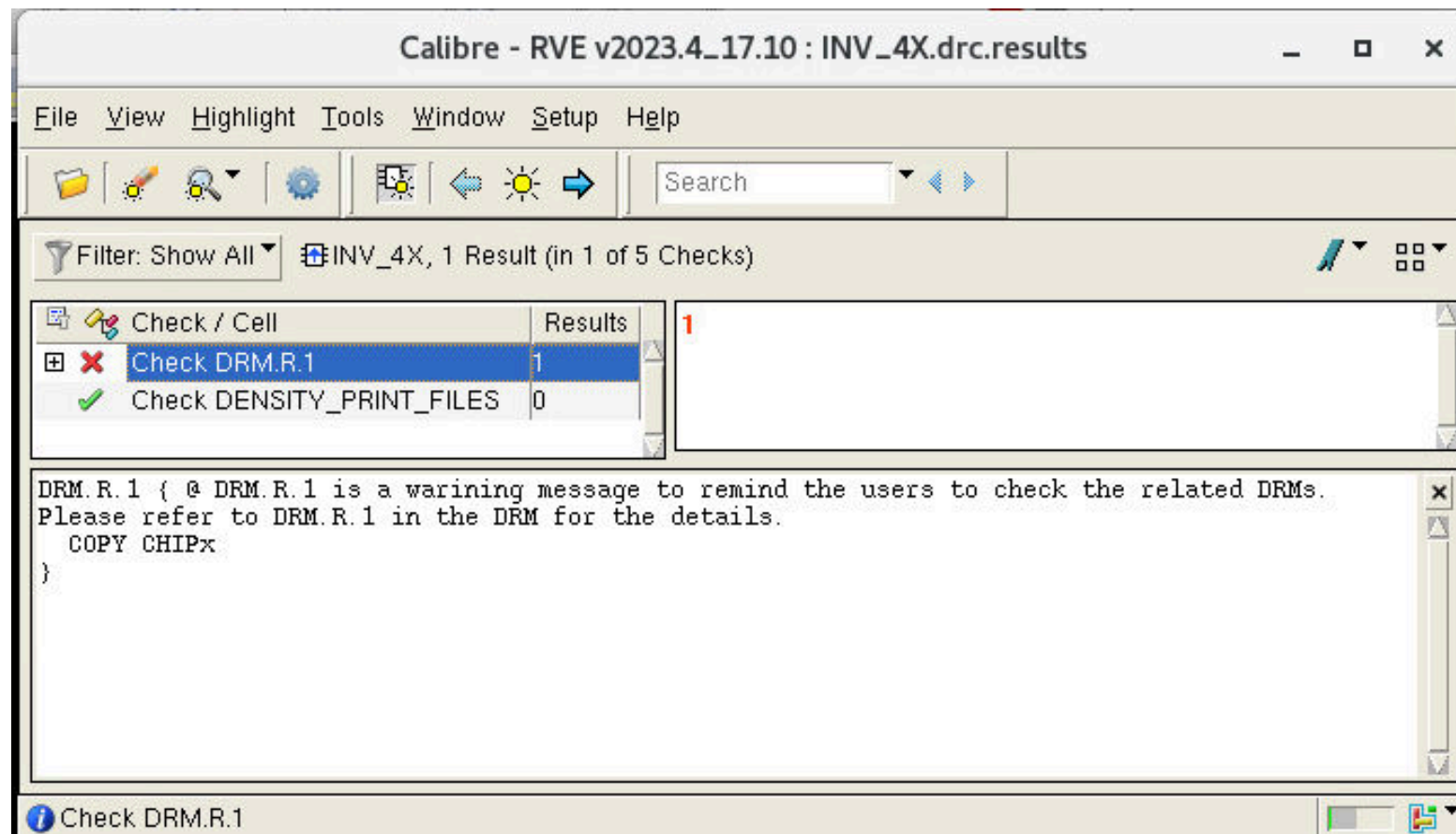
5.2、Oscillator



◆ INV_4X版图;



1、INV_4X layout;

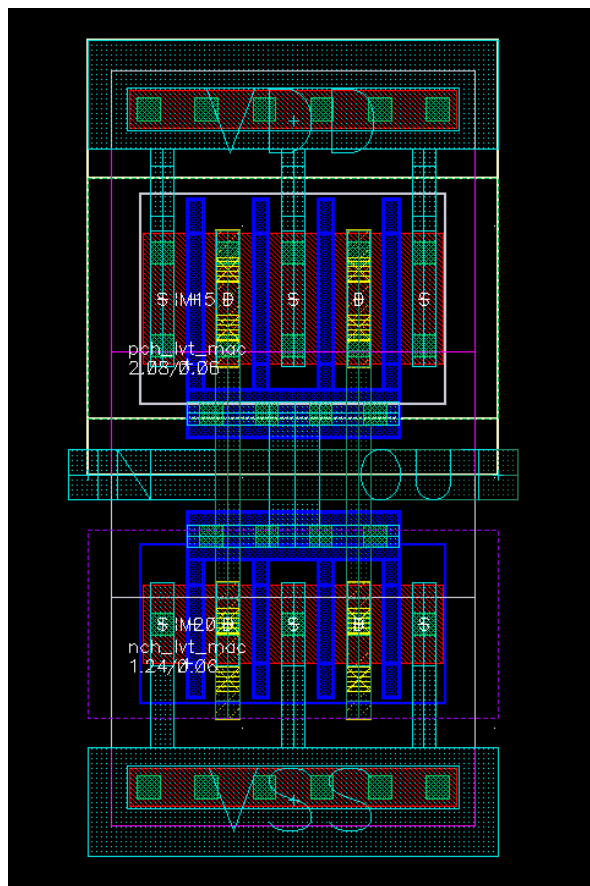


2、INV_4X layout **DRC** Results;

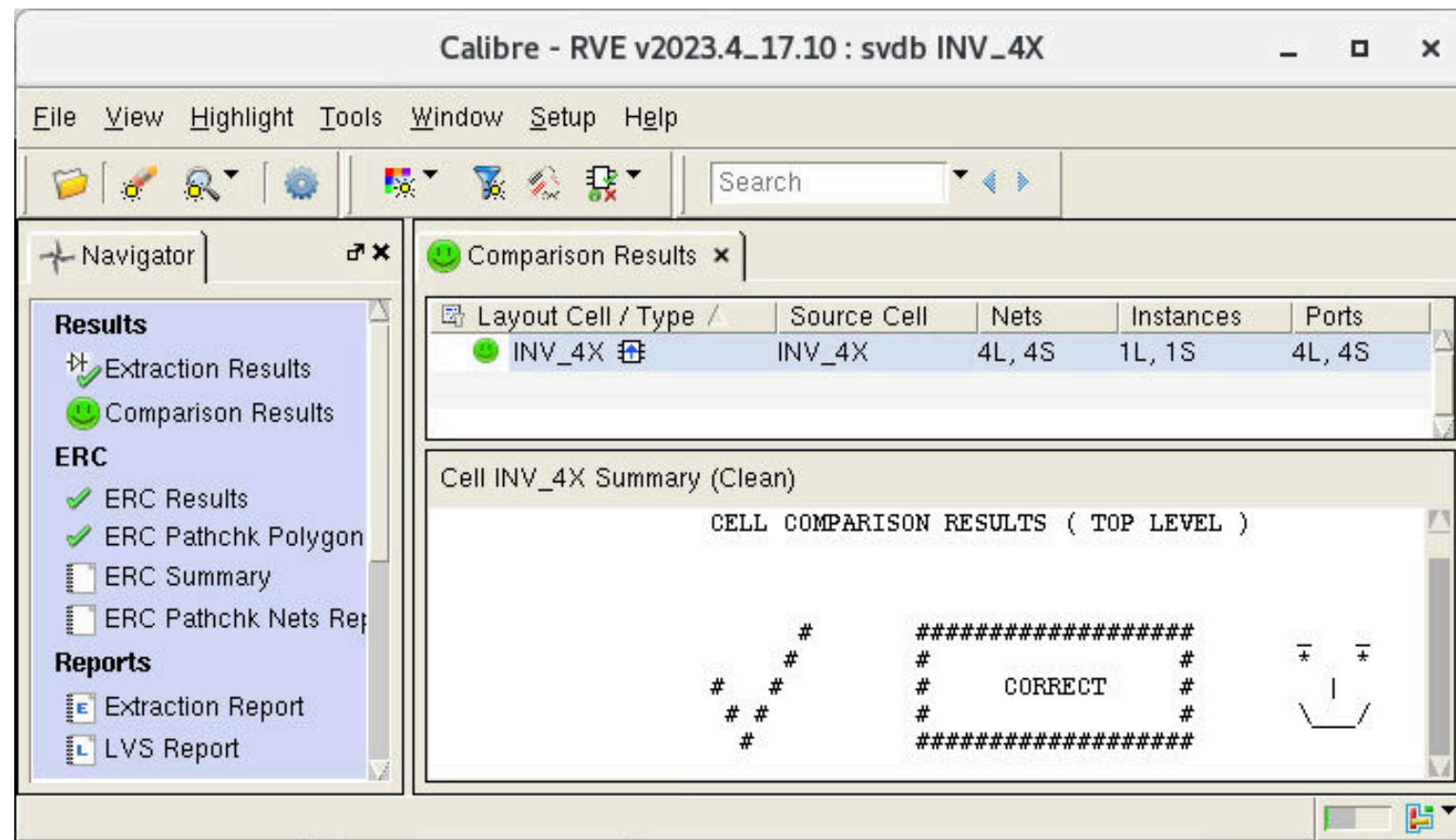
5.2、Oscillator



◆ INV_4X版图;



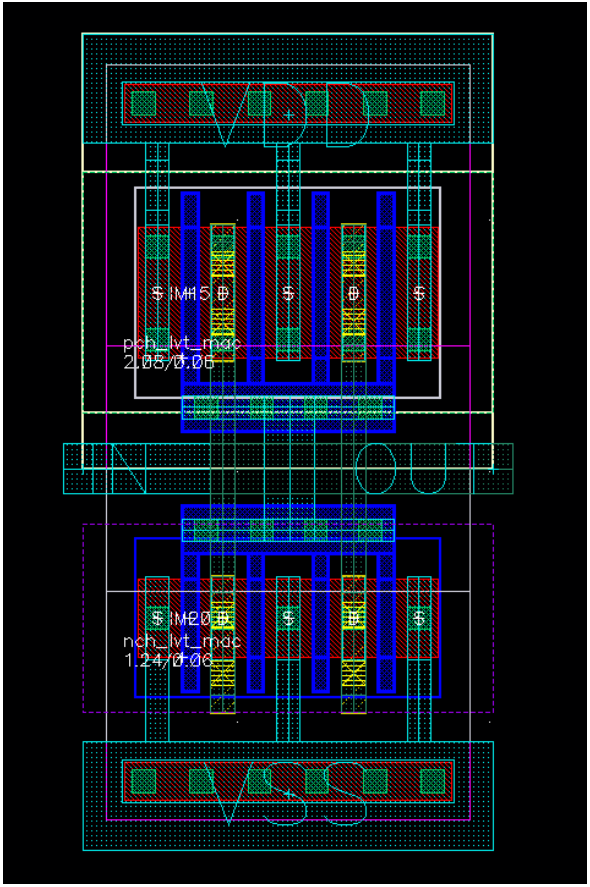
1、INV_4X layout;



2、INV_4X layout **LVS** Results;

5.2、Oscillator

◆ INV_4X版图;



Calibre - RVE v2023.4.17.10 : svdb INV_4X

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INV_4X

- No. Layout Net Source Net R Count C Total CC Total C+CC Total
- 1 VSS VSS 28 1.78043E-16 3.80769E-16 5.58812E-16
- 2 VDD VDD 24 1.82460E-16 4.69711E-16 6.52172E-16
- 3 OUT OUT 37 7.09536E-17 8.82703E-16 9.53656E-16
- 4 IN IN 53 2.44453E-16 7.89235E-16 1.03369E-15

Find Nets: Type in nets to search for

Coupling to: All Nets Specified Nets Type in specific nets for coupling

IN

- Type Count Total Source
- R 53
- Pt-to-Pt Res. 0
- C 26 2.44453E-16
- CC 111 7.89235E-16

Layout Net: IN Source Net: IN Reduction Occurred Resistors: R (53 R)

No.	Layer	Value	Property
1	metal1	0.268120	w=0.200000, thickness=1.85696E-07, awidth=0.200000, connect_rule_num=1.00...
2	metal1	0.237778	w=0.200000, thickness=1.85696E-07, awidth=0.200000, connect_rule_num=1.00...
3	metal1	0.00366739	w=0.200000, thickness=1.85696E-07, awidth=0.200000, connect_rule_num=1.00...
4	metal1	0.144110	w=0.090000, thickness=1.80985E-07, awidth=0.090000, connect_rule_num=1.00...
5	metal1	0.0748134	w=0.090000, thickness=1.85696E-07, awidth=0.090000, connect_rule_num=1.00...
6	metal1	0.0187970	w=0.090000, thickness=1.80985E-07, awidth=0.090000, connect_rule_num=1.00...
7	metal1	0.151716	w=0.090000, thickness=1.80985E-07, awidth=0.090000, connect_rule_num=1.00...
8	metal1	0.144110	w=0.090000, thickness=1.85696E-07, awidth=0.090000, connect_rule_num=1.00...
9	metal1	0.0748134	w=0.090000, thickness=1.85696E-07, awidth=0.090000, connect_rule_num=1.00...
10	metal1	0.0187970	w=0.090000, thickness=1.80985E-07, awidth=0.090000, connect_rule_num=1.00...
11	metal1	0.151716	w=0.090000, thickness=1.80985E-07, awidth=0.090000, connect_rule_num=1.00...
12	metal1	0.00489126	w=0.090000, thickness=1.85696E-07, awidth=0.090000, connect_rule_num=1.00...
13	metal1	0.0748134	w=0.090000, thickness=1.80985E-07, awidth=0.090000, connect_rule_num=1.00...
14	metal1	0.00489126	w=0.090000, thickness=1.85696E-07, awidth=0.090000, connect_rule_num=1.00...
15	metal1	0.0748134	w=0.090000, thickness=1.85696E-07, awidth=0.090000, connect_rule_num=1.00...
16	metal1	0.178707	w=0.200000, thickness=1.85696E-07, awidth=0.200000, connect_rule_num=1.00...
17	metal1	0.178707	w=0.200000, thickness=1.85696E-07, awidth=0.200000, connect_rule_num=1.00...
18	poly	65.0250	w=0.060000, thickness=1.00000E-07, awidth=0.060000, connect_rule_num=1.00...
19	poly	65.0250	w=0.060000, thickness=1.00000E-07, awidth=0.060000, connect_rule_num=1.00...
20	poly	65.0250	w=0.060000, thickness=1.00000E-07, awidth=0.060000, connect_rule_num=1.00...
21	poly	65.0250	w=0.060000, thickness=1.00000E-07, awidth=0.060000, connect_rule_num=1.00...
22	poly	91.8000	w=0.060000, thickness=1.00000E-07, awidth=0.060000, connect_rule_num=1.00...
23	poly	91.8000	w=0.060000, thickness=1.00000E-07, awidth=0.060000, connect_rule_num=1.00...
24	poly	91.8000	w=0.060000, thickness=1.00000E-07, awidth=0.060000, connect_rule_num=1.00...
25	poly	91.8000	w=0.060000, thickness=1.00000E-07, awidth=0.060000, connect_rule_num=1.00...
26	poly	11.9758	w=0.190000, thickness=1.00000E-07, awidth=0.190000, connect_rule_num=1.00...
27	poly	11.9758	w=0.190000, thickness=1.00000E-07, awidth=0.190000, connect_rule_num=1.00...
28	poly	13.2868	w=0.190000, thickness=1.00000E-07, awidth=0.190000, connect_rule_num=1.00...
29	poly	2.29487	w=0.190000, thickness=1.00000E-07, awidth=0.190000, connect_rule_num=1.00...

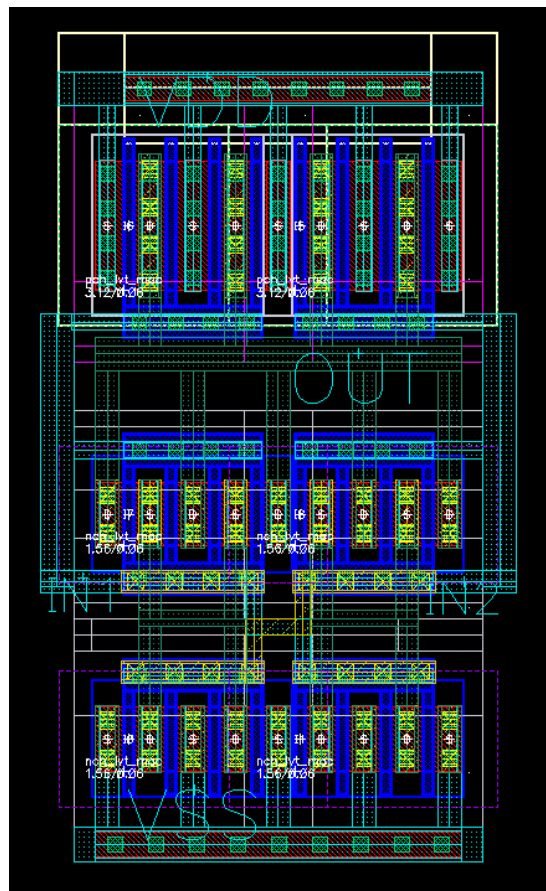
1、INV_4X layout;

2、INV_4X layout PEX Results;

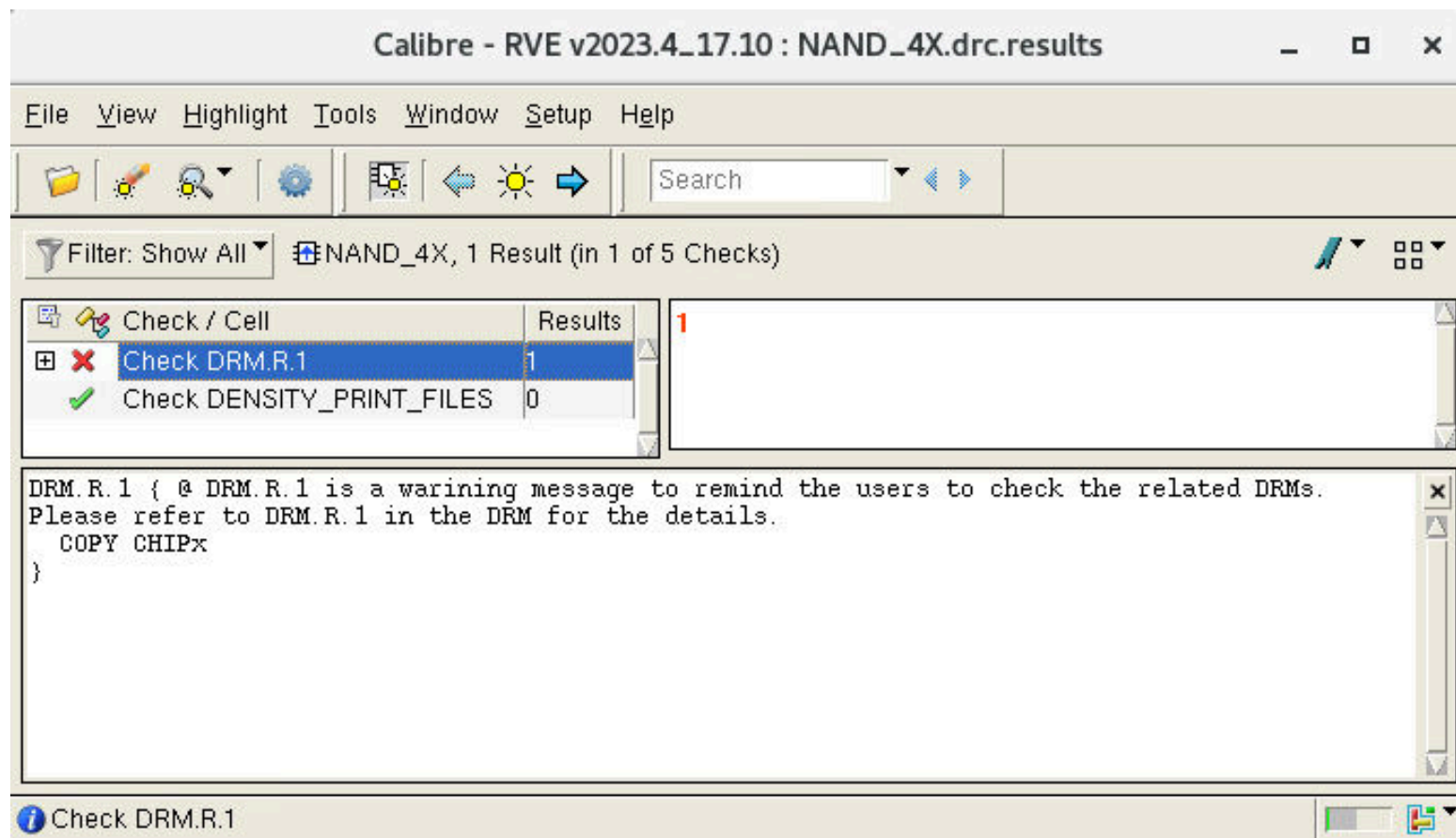
5.2、Oscillator



◆ NAND_4X版图;



1、NAND_4X layout;

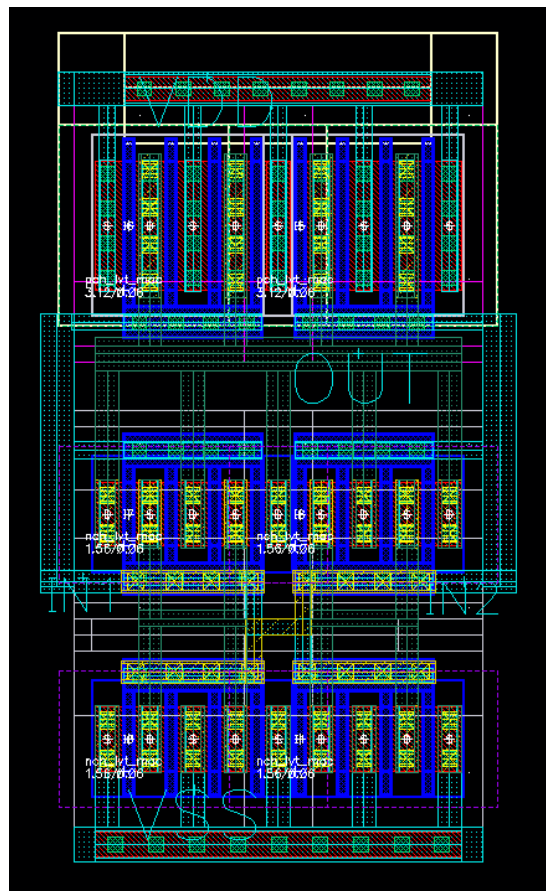


2、NAND_4X layout **DRC** Results;

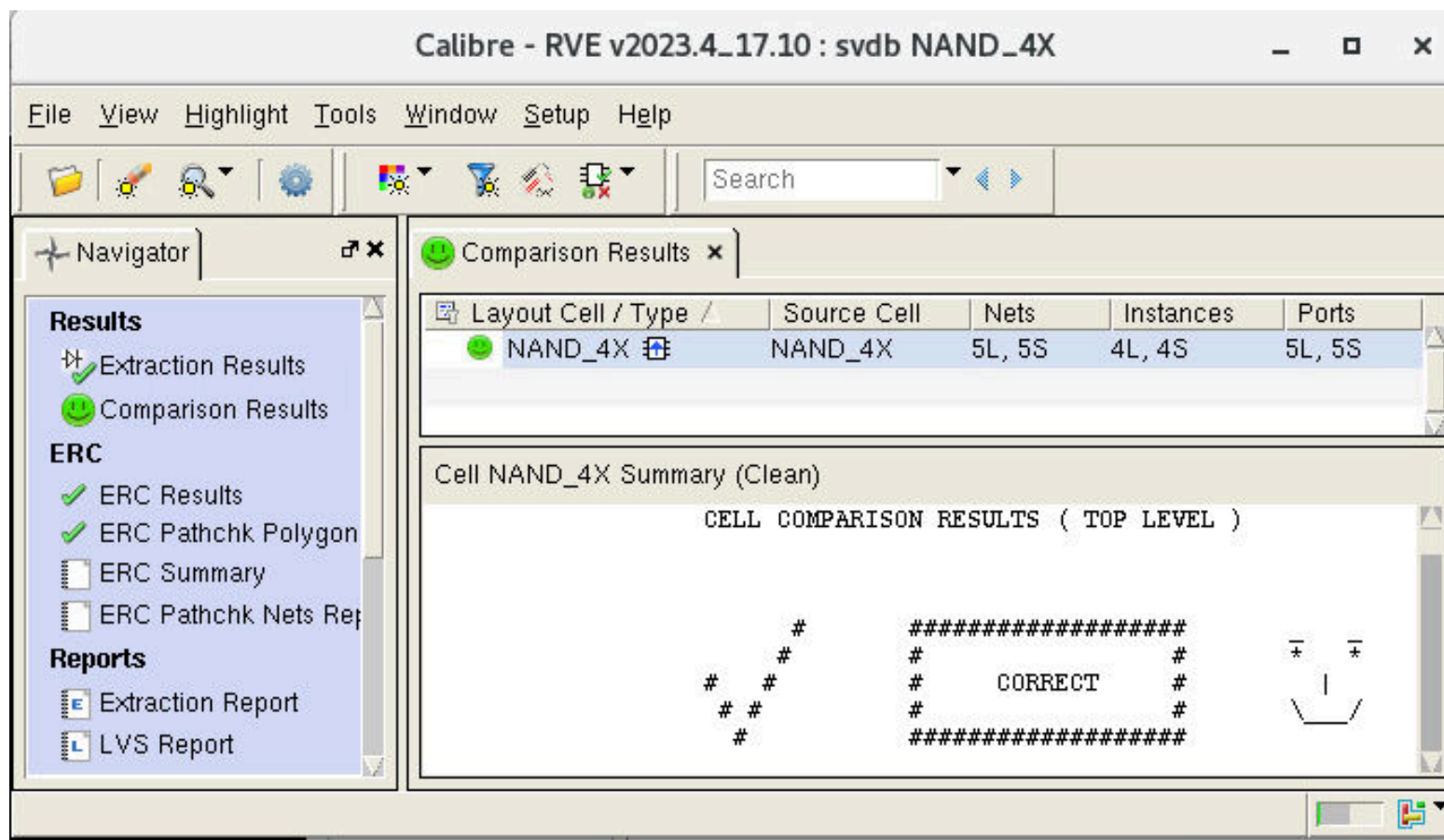
5.2、Oscillator



◆ NAND_4X版图;



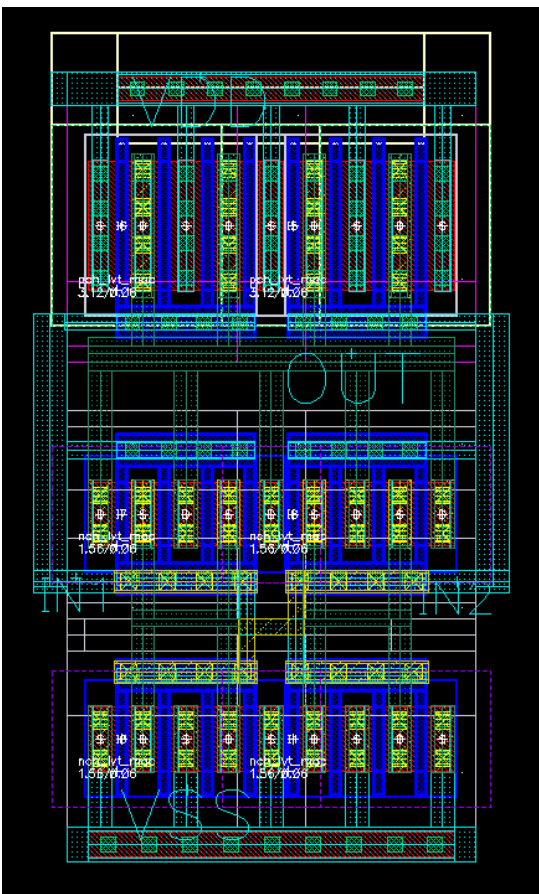
1、NAND_4X layout;



2、NAND_4X layout **LVS** Results;

5.2、Oscillator

◆ NAND_4X版图;



Calibre - RVE v2023.4.17.10 : svdb NAND_4X

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NAND_4X

No.	Layout Net	Source Net	R Count	C Total (F)	CC Total (F)	C+CC Total (F)
1	VSS	VSS	47	2.10438E-16	1.41082E-15	1.62126E-15
2	VDD	VDD	59	2.12710E-16	1.41387E-15	1.62658E-15
3	OUT	OUT	112	1.34610E-16	3.02800E-15	3.16261E-15
4	4	NET1	43	4.32615E-17	1.55811E-15	1.60137E-15
5	5	NET2	43	4.30591E-17	1.55871E-15	1.60177E-15
6	IN2	IN2	141	4.74550E-16	2.27044E-15	2.74499E-15
7	IN1	IN1	144	4.92775E-16	2.25202E-15	2.74479E-15

Find Nets: Type in nets to search for

OUT

Type	Count	Total	Source
R	112		
PT-to-PT Res.	0		
C	28	1.34610E-16	
CC	310	3.02800E-15	
4	45	4.33228E-16	NET1
5	45	4.33585E-16	NET2
IN1	80	7.40905E-16	IN1
IN2	79	7.38359E-16	IN2
VDD	56	6.77279E-16	VDD
VSS	5	4.64214E-16	VSS

OUT

No.	Layer	Value	Property
1	metal1	0.00623558	w=0.140000, thickness=1.82913E-07, awidth=0.140000, connect_rule_num=1.00...
2	metal1	0.00623558	w=0.140000, thickness=1.82913E-07, awidth=0.140000, connect_rule_num=1.00...
3	metal1	0.00632406	w=0.140000, thickness=1.82913E-07, awidth=0.140000, connect_rule_num=1.00...
4	metal1	0.00632406	w=0.140000, thickness=1.82913E-07, awidth=0.140000, connect_rule_num=1.00...
5	metal1	0.00632406	w=0.140000, thickness=1.82913E-07, awidth=0.140000, connect_rule_num=1.00...
6	metal1	0.425527	w=0.140000, thickness=1.82913E-07, awidth=0.140000, connect_rule_num=1.00...
7	metal1	0.0303948	w=0.140000, thickness=1.82913E-07, awidth=0.140000, connect_rule_num=1.00...
8	metal1	0.0101316	w=0.140000, thickness=1.82913E-07, awidth=0.140000, connect_rule_num=1.00...
9	metal1	0.425527	w=0.140000, thickness=1.82913E-07, awidth=0.140000, connect_rule_num=1.00...
10	metal1	0.0303948	w=0.140000, thickness=1.82913E-07, awidth=0.140000, connect_rule_num=1.00...
11	metal1	0.0101316	w=0.140000, thickness=1.82913E-07, awidth=0.140000, connect_rule_num=1.00...
12	metal1	0.425527	w=0.140000, thickness=1.82913E-07, awidth=0.140000, connect_rule_num=1.00...
13	metal1	0.0303948	w=0.140000, thickness=1.82913E-07, awidth=0.140000, connect_rule_num=1.00...
14	metal1	0.0101316	w=0.140000, thickness=1.82913E-07, awidth=0.140000, connect_rule_num=1.00...
15	metal1	0.425527	w=0.140000, thickness=1.82913E-07, awidth=0.140000, connect_rule_num=1.00...
16	metal1	0.0303948	w=0.140000, thickness=1.82913E-07, awidth=0.140000, connect_rule_num=1.00...
17	metal1	0.0101316	w=0.140000, thickness=1.82913E-07, awidth=0.140000, connect_rule_num=1.00...
18	metal2	0.331711	w=0.140000, thickness=2.30628E-07, awidth=0.140000, connect_rule_num=1.00...
19	metal2	0.331711	w=0.140000, thickness=2.30628E-07, awidth=0.140000, connect_rule_num=1.00...
20	metal2	0.331711	w=0.140000, thickness=2.30628E-07, awidth=0.140000, connect_rule_num=1.00...
21	metal2	0.331711	w=0.140000, thickness=2.30628E-07, awidth=0.140000, connect_rule_num=1.00...
22	metal3	0.257040	w=0.140000, thickness=2.28320E-07, awidth=0.140000, connect_rule_num=1.00...
23	metal3	0.257040	w=0.140000, thickness=2.28320E-07, awidth=0.140000, connect_rule_num=1.00...
24	metal3	0.286339	w=0.140000, thickness=2.28320E-07, awidth=0.140000, connect_rule_num=1.00...
25	metal3	0.286339	w=0.140000, thickness=2.28320E-07, awidth=0.140000, connect_rule_num=1.00...
26	metal3	0.286339	w=0.140000, thickness=2.28320E-07, awidth=0.140000, connect_rule_num=1.00...
27	metal3	0.143693	w=0.200000, thickness=2.30967E-07, awidth=0.200000, connect_rule_num=1.00...
28	metal3	0.0223523	w=0.200000, thickness=2.30967E-07, awidth=0.200000, connect_rule_num=1.00...
29	metal3	0.488028	w=0.140000, thickness=2.28320E-07, awidth=0.140000, connect_rule_num=1.00...

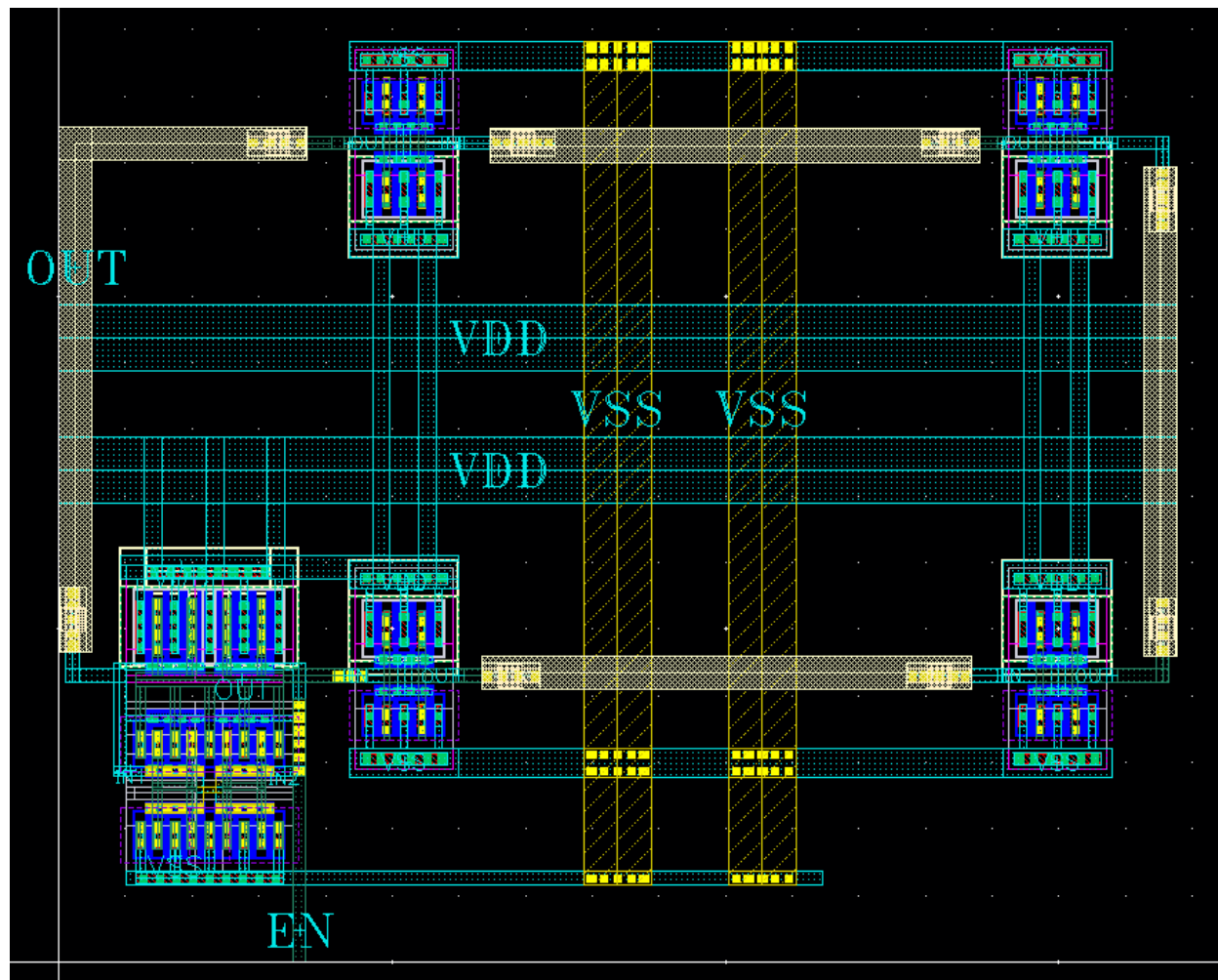
1、NAND_4X layout;

2、NAND_4X layout PEX Results;

5.2、Oscillator



◆ Oscillator版图;

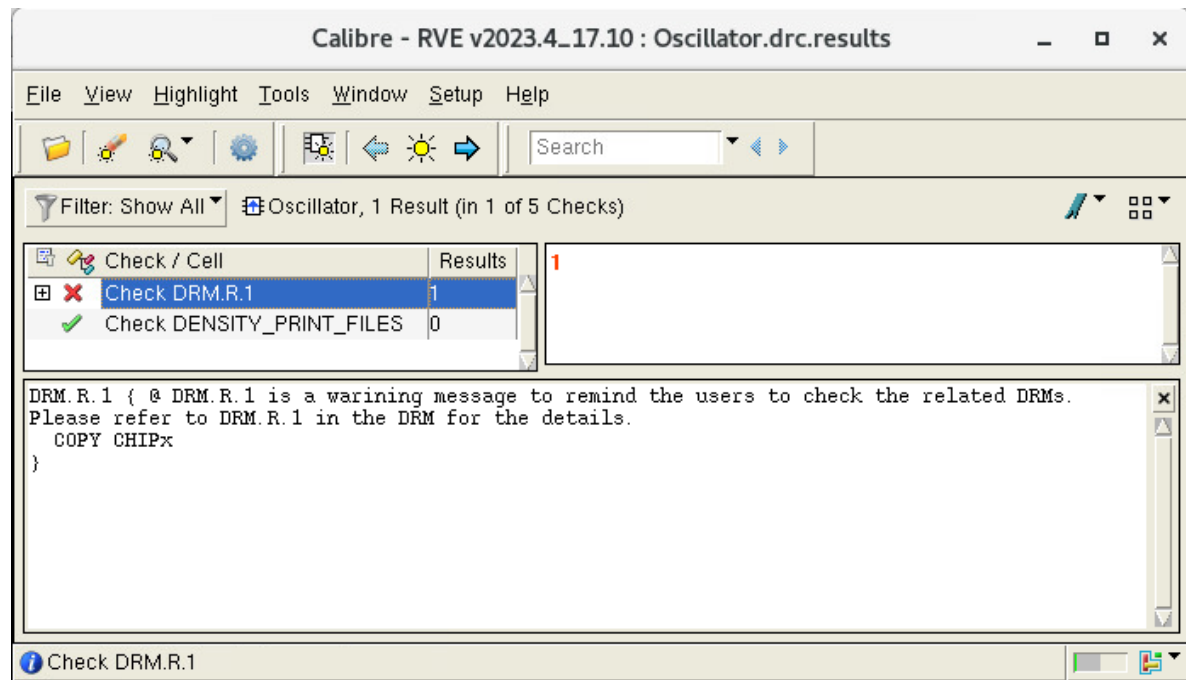


1、Oscillator layout;

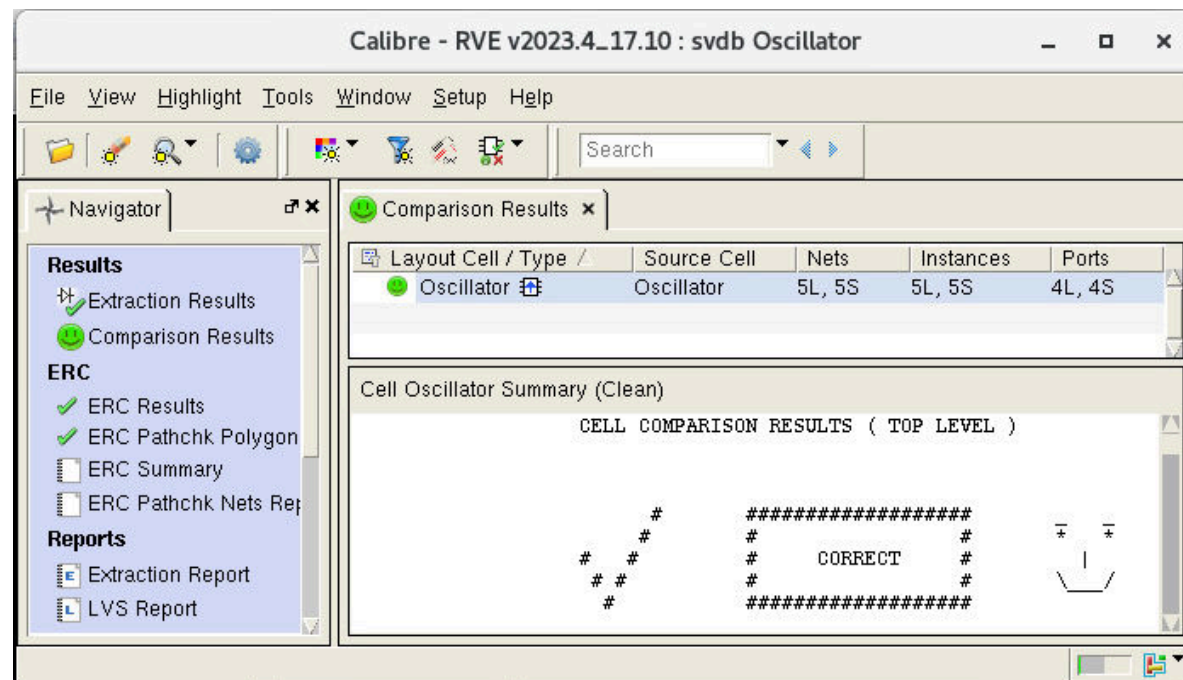


5.2、Oscillator

◆ Oscillator版图;



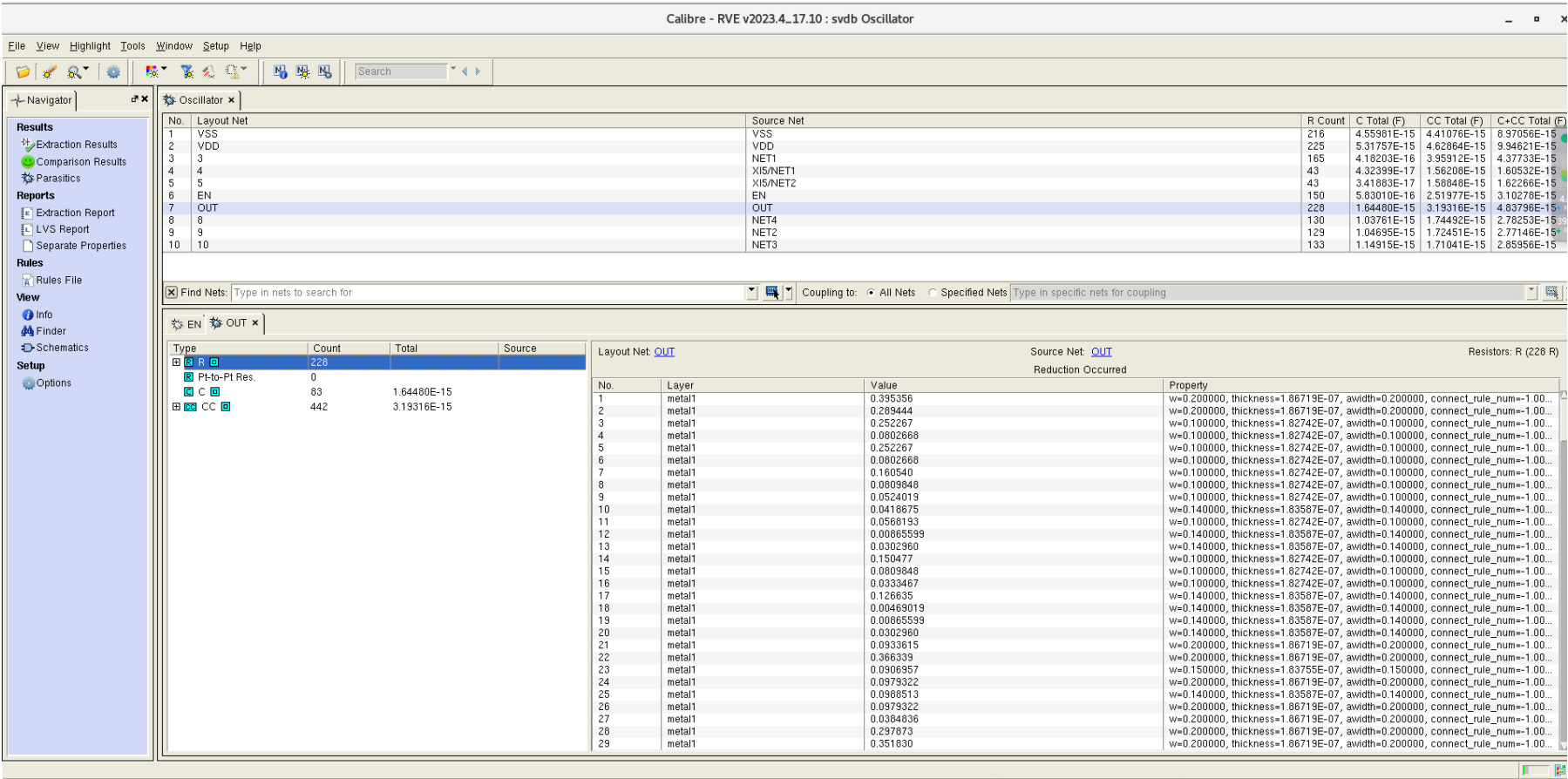
1、Oscillator layout **DRC** Results;



2、Oscillator layout **LVS** Results;

5.2、Oscillator

◆ Oscillator版图;

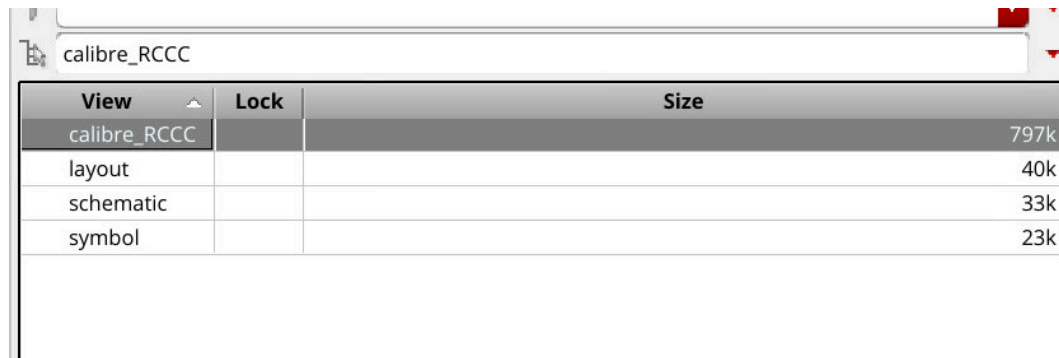


1、Oscillator layout PEX Results;

5.2、Oscillator输出波形仿真——后仿

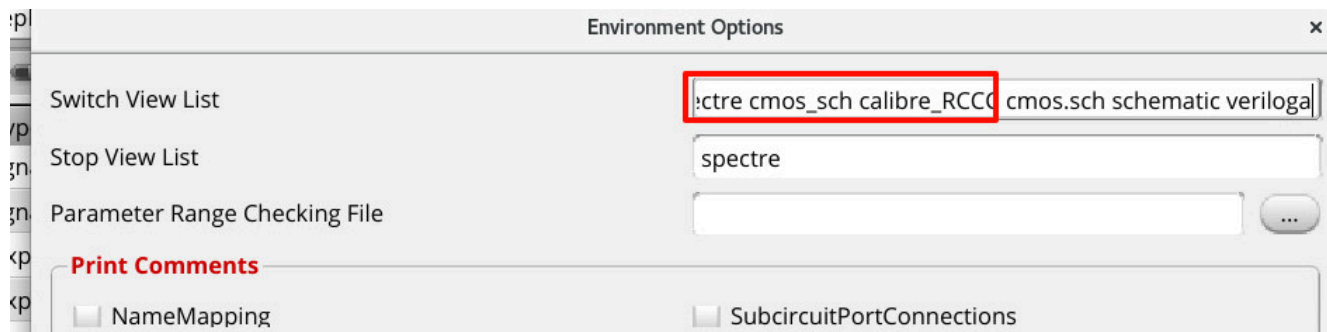


确认 Cell view 下有 Calibre_RCCC 寄生文件



View	Lock	Size
calibre_RCCC		797k
layout		40k
schematic		33k
symbol		23k

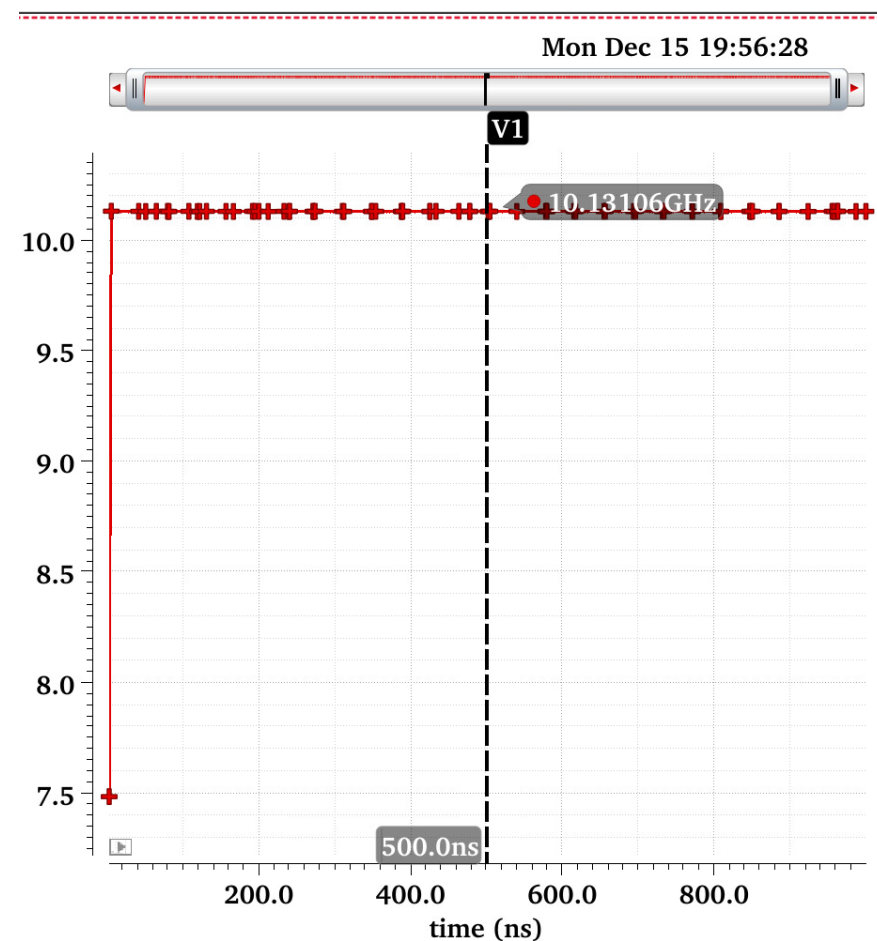
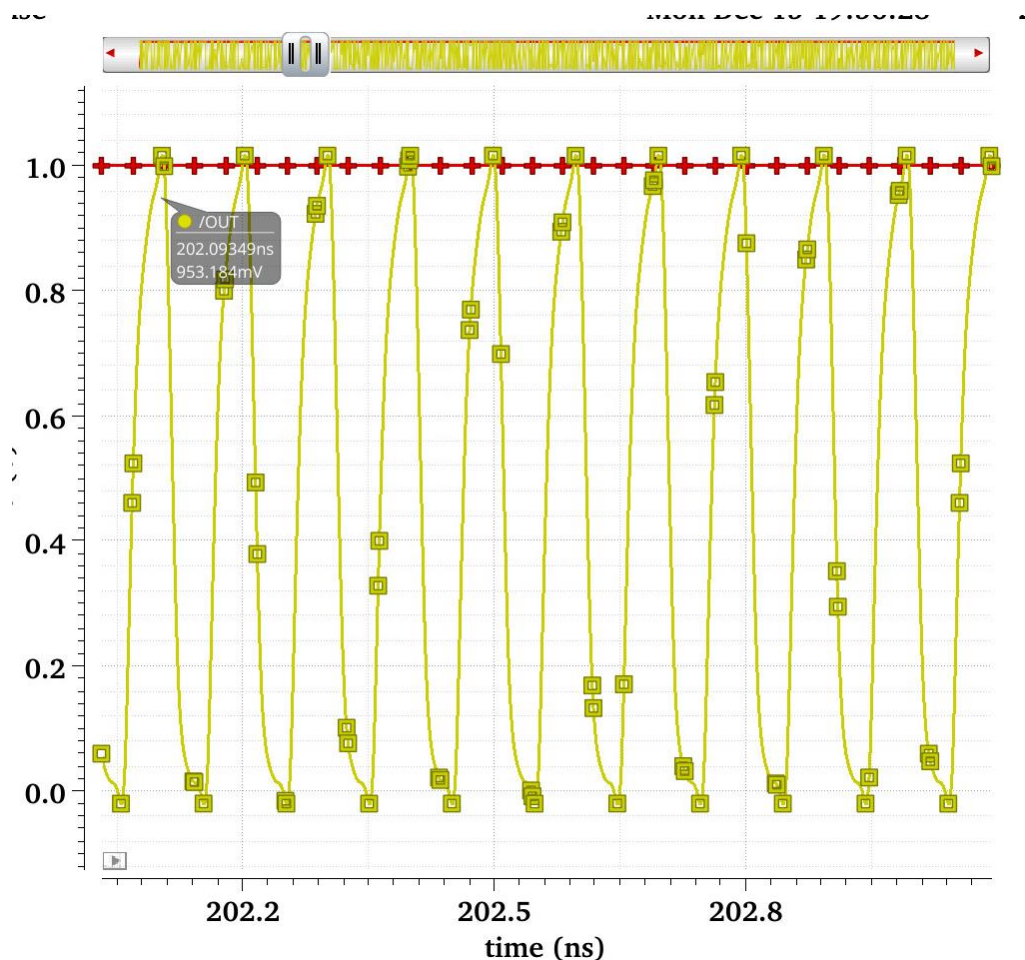
Maestro 页面中, setup -> Environment Options 中添加 RCCC寄生文件, 其余步骤与前仿一致。



5.2、Oscillator输出波形仿真——后仿



振荡器波形输出，频率从17GHz降低到10GHz





5.2、Oscillator Corner输出波形仿真——后仿

振荡器输出频率, corner 降低了10GHz

Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
corner_0	0	900m	-20	ff_lvt					11.58G
corner_1	0	900m	25	ff_lvt					11.13G
corner_2	0	900m	85	ff_lvt					10.64G
corner_3	0	1	-20	ff_lvt					13.21G
corner_4	0	1	25	ff_lvt					12.6G
corner_5	0	1	85	ff_lvt					11.89G
corner_6	0	1.1	-20	ff_lvt					14.67G

Corner	v_P1	v_VDD	temperature	cor_lvt.scs	Pass/Fail	/EN	w_out	w_freq	..." ?threshold 0.0) 5e-07)
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
nom	0	1	27	<omitted>					17.76G
corner_0	0	900m	-20	ff_lvt					20.46G
corner_1	0	900m	25	ff_lvt					19.33G
corner_2	0	900m	85	ff_lvt					18.12G
corner_3	0	1	-20	ff_lvt					23.21G
corner_4	0	1	25	ff_lvt					21.75G
corner_5	0	1	85	ff_lvt					20.2G
corner_6	0	1.1	-20	ff_lvt					25.63G



谢谢大家！